Inferential Uncertainty in RNA-Seq and its downstream applications

Recap

In the last lecture we covered:

- RNA-Seq quantification problem
- Generative model for RNA-Seq
- (Range-factorized) Equivalence classes
- Salmon
- Estimating Posterior Uncertainty

Quantifying Uncertainty

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Nonparametric expression analysis using inferential replicate counts

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Inferential relative variance

$$\mathrm{infRV}_{\mathbf{g_i}} = \frac{\max(0, \mathrm{var}(\mathbf{g_i}) - \mathrm{mean}(\mathbf{g_i}))}{\mathrm{mean}(\mathbf{g_i}) + p} + s$$

Why (variance - mean) / mean?

Minimally within a sample we have Poisson variance which scales with the mean.

So InfRV = ~0 with no multi-mapping.

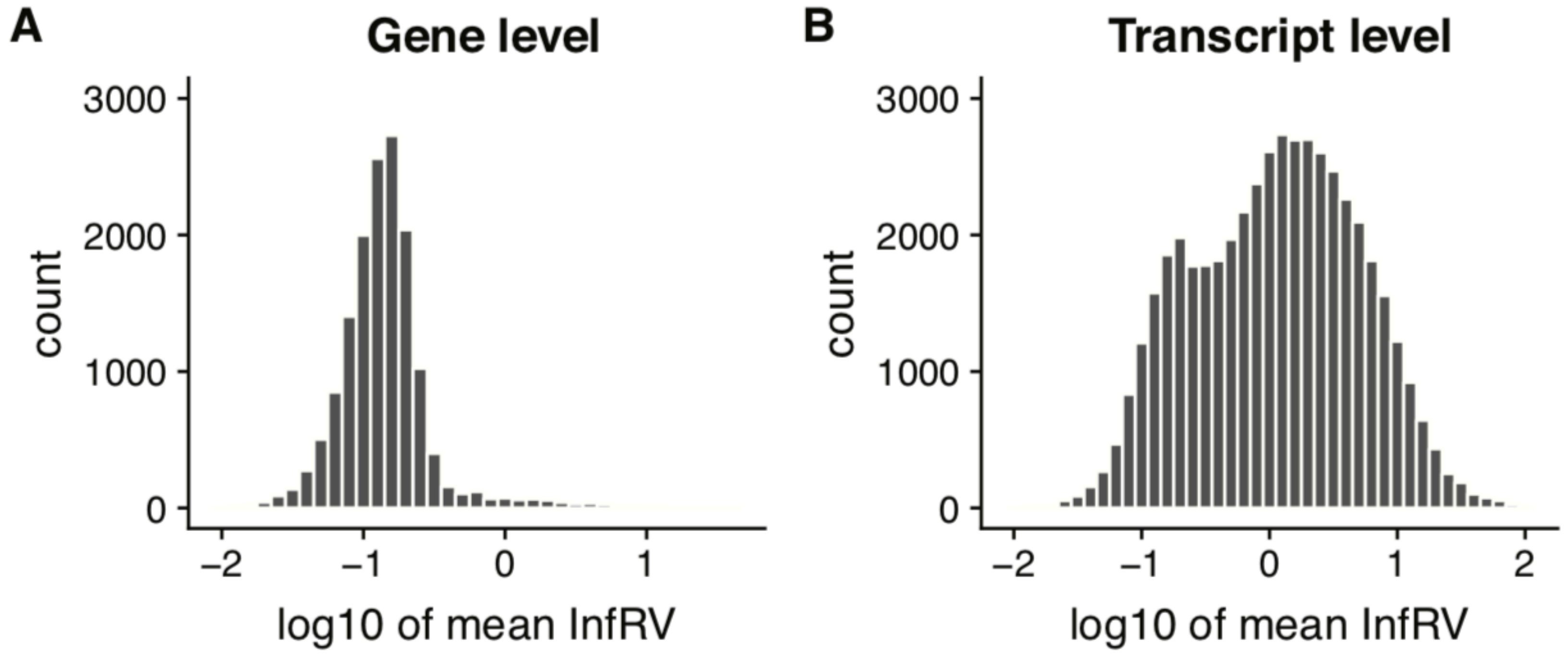
Larger the value of InfRV, more uncertainty

Features for most downstream analysis

- People already perform analyses that greatly reduce such uncertainty; gene-level analysis

- But, gene-level analysis is often *too* coarse-grained.

- Many transcripts can be confidently quantified; some cannot; which transcripts are “impossible” differs from sample-to-sample.



We need “sub-gene” level transcriptional resolution

How should we, in general, address this problem?

What is the computational challenge here?

Let the data, themselves, tell you what the “inferentially distinguishable” transcriptional groups should be!

Using posterior uncertainty as a signal

As with MMSeq, Turro had an elegant solution to this as well!

Gene expression Advance Access publication November 26, 2013

Flexible analysis of RNA-seq data using mixed effects models

Ernest Turro[1,2,*], William J. Astle[3] and Simon Tavaré[1]

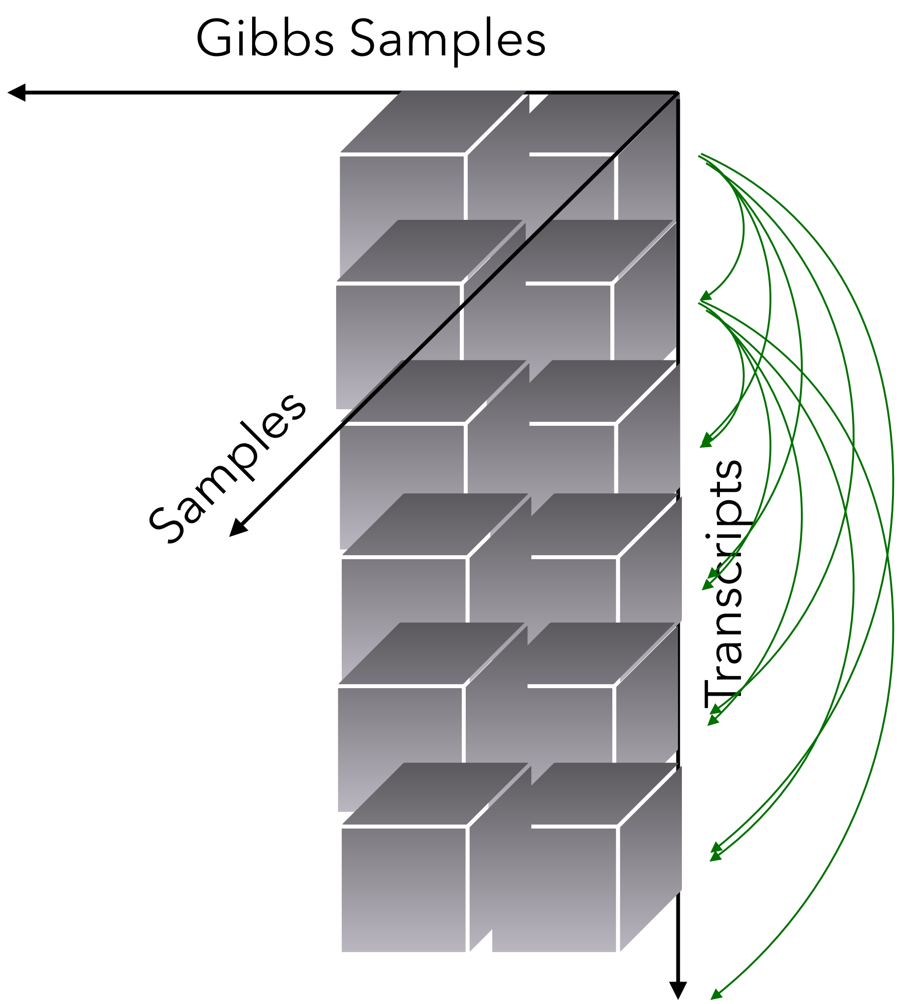
[1]Cancer Research UK Cambridge Institute, University of Cambridge, Robinson Way, Cambridge CB2 0RE, UK, [2]Department of Haematology, University of Cambridge, NHS Blood and Transplant, Long Road, Cambridge CB2 0PT, UK and [3]Department of Epidemiology, Biostatistics and Occupational Health, McGill University, 1020 Pine Avenue West, Montreal QC H3A 1A2, Canada

Associate Editor: Ivo Hofacker

> “ Often, sets of transcripts with poorly estimated expression parameters are anti-correlated because reads can be mapped to the combined set of transcripts with confidence, but no reads can be mapped to specific transcripts within the set. In these circumstances, it may be more informative to treat the set of transcripts as the unit of inference, rather than the individual transcripts.
>
> We now propose an algorithm for collapsing transcripts with anti-correlated expression parameters into informative sets. “

Using posterior uncertainty as a signal

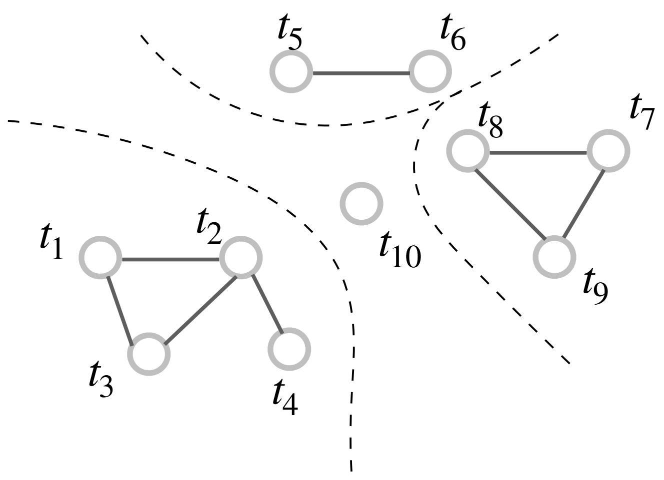
Unfortunately, this implementation tends to be impractical as transcriptome complexity increases, and also discovered some shortcomings with the particular approach.



mmcollapse

$$\binom{n}{2} \times m$$

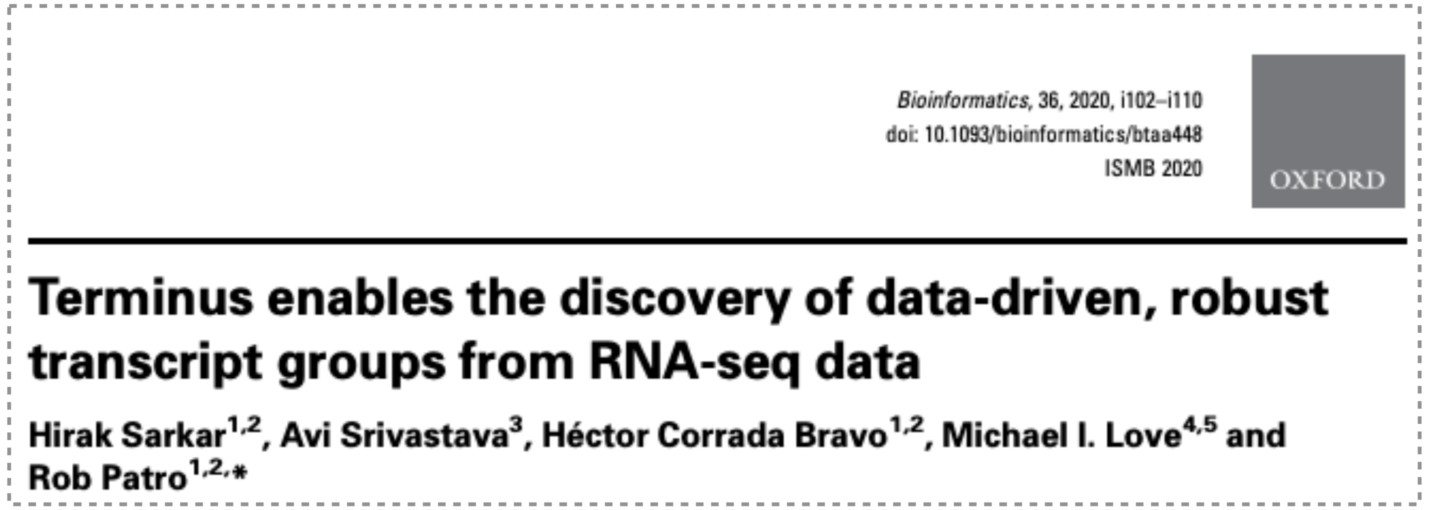
This many correlation calculation at the first step



Collapsing would take place constrained by the ***fragment ambiguity graph***

terminus

Terminus: Discovery of Data-driven Transcriptional Groups


Bioinformatics, 36, 2020, i102–i110
doi: 10.1093/bioinformatics/btaa448
ISMB 2020

OXFORD

Terminus enables the discovery of data-driven, robust transcript groups from RNA-seq data

Hirak Sarkar[1,2], Avi Srivastava[3], Héctor Corrada Bravo[1,2], Michael I. Love[4,5] and Rob Patro[1,2,*]

Free & open-source tool (BSD3), available at :

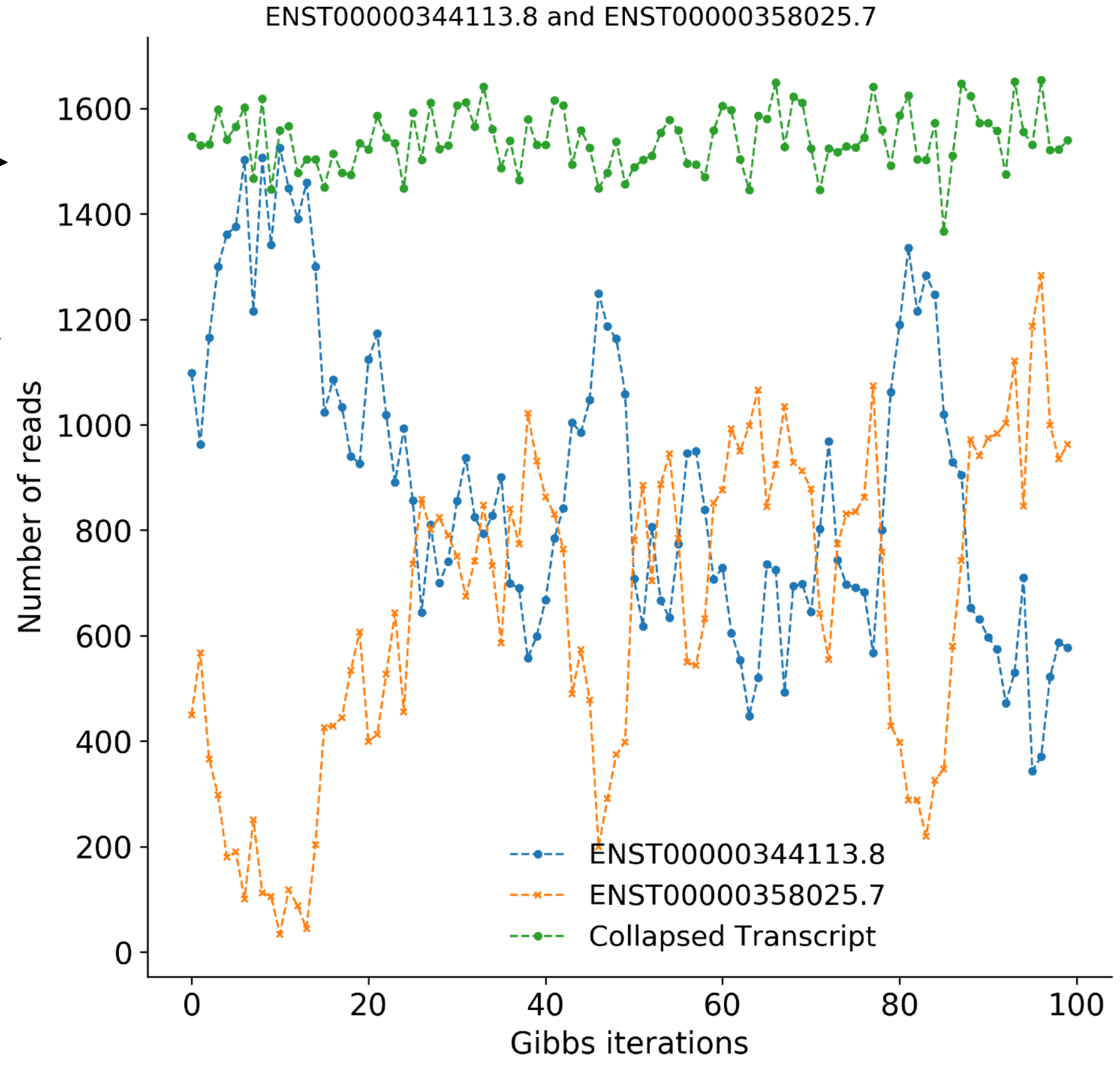
https://github.com/COMBINE-lab/Terminus

Defining transcriptional groups

Goal : Find groups of transcripts that, independently, have large inferential uncertainty, but, when grouped together, can be confidently quantified.

Actual Gibbs traces for a pair of transcripts merged by our algorithm.

Actual Gibbs traces for a pair of transcripts merged by our algorithm.



A metric for choosing transcripts to group

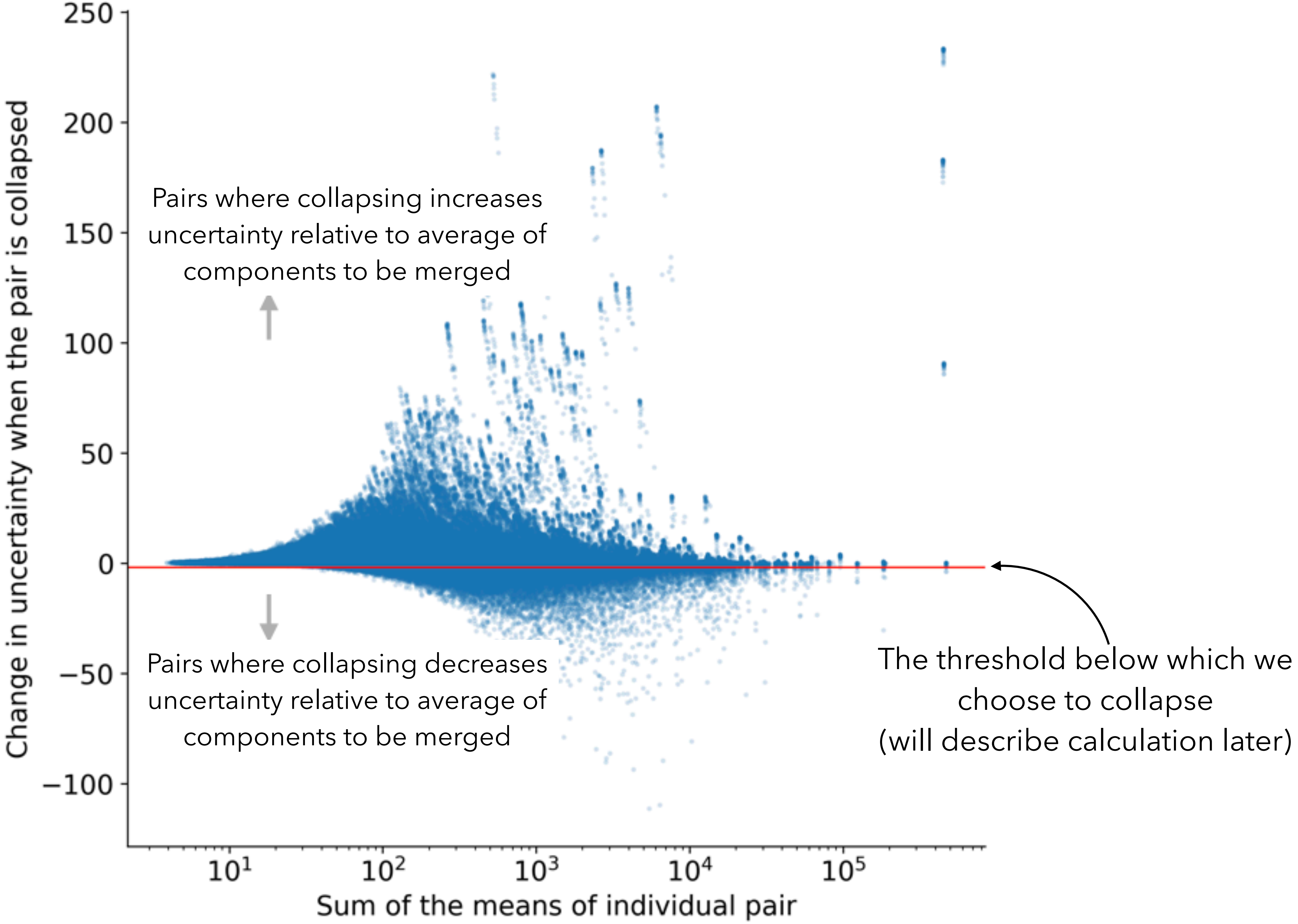
- We will employ an iterative algorithm that collapses groups that are “good candidates”; what makes a "good candidate”?

- We define the change of $InfRV$ between the transcriptional group and average of individual group members as a metric to identify the promising transcripts to merge.

- For two transcripts (or previously defined groups) v_i and v_j we define the *score* for merging them as $InfRV$ as :

$$s(v_i, v_j) = infRV(g_i + g_j) - \frac{infRV(g_i) + infRV(g_j)}{2}$$

How does this metric look?

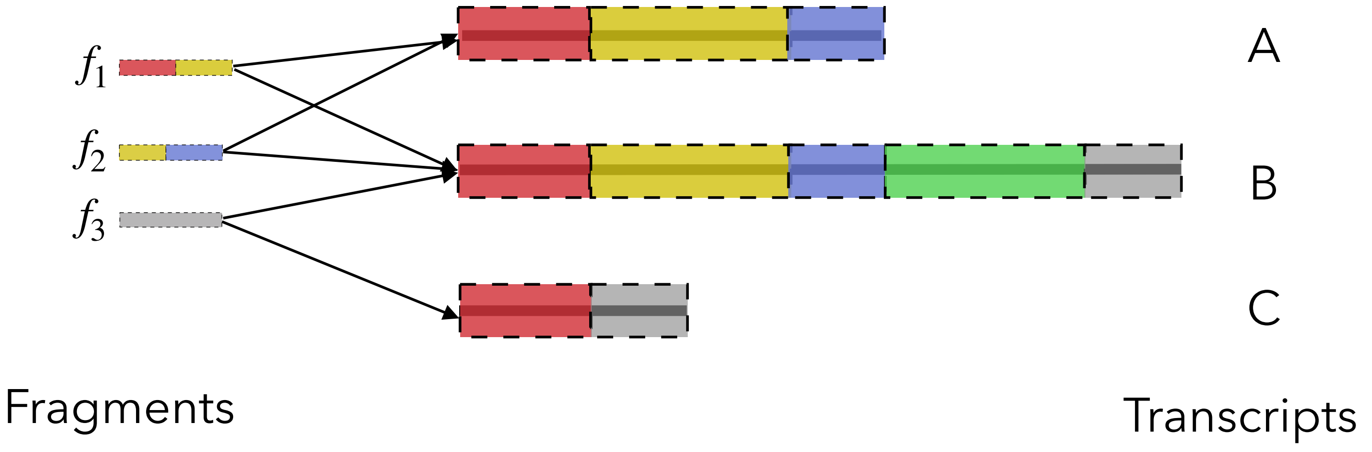


Evaluating all potential candidates is (far) too slow!

- For T transcripts there are $\binom{T}{2}$ pairs that should be considered.
- Not all pairs are important. In other words, almost all pairs should *not* be good candidates for collapse
- Idea: Use structure of alignment ambiguity to efficiently cull the search space!

Fragment Equivalence Classes and the Fragment Ambiguity Graphs

Mapping Information



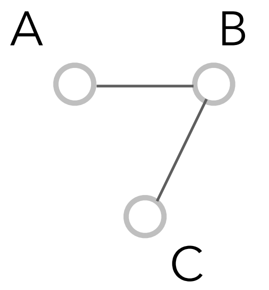
Equivalence Relation

$f_1 \longrightarrow$ (A, B)
$f_2 \longrightarrow$ (A, B)
$f_3 \longrightarrow$ (B, C)

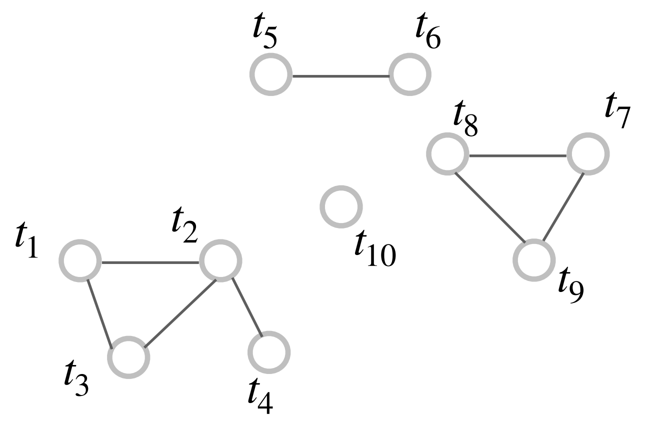
Fragments   Equivalence Class

(A, B) $\longrightarrow$ 2
(B, C) $\longrightarrow$ 1

Network



The resulting network (graph) describes the "fragment-level" ambiguity



$$\mathscr{E}_j : \{t_5, t_6\} \rightarrow \{\underline{w_{5,j}, w_{6,j}}; c\}$$

Note: An edge can be labeled with > 1 equivalence class.

Rich Equivalence Class Attributes

- Equivalence Class Count
- Conditional probabilities $w_{i,j}$ for transcript i in equivalence class j. It consolidates the probability of $\sum_{f_k \in \mathscr{E}_j} P\{f_k \mid t_i\}$

Fragment Equivalence Classes and the Fragment Ambiguity Graphs

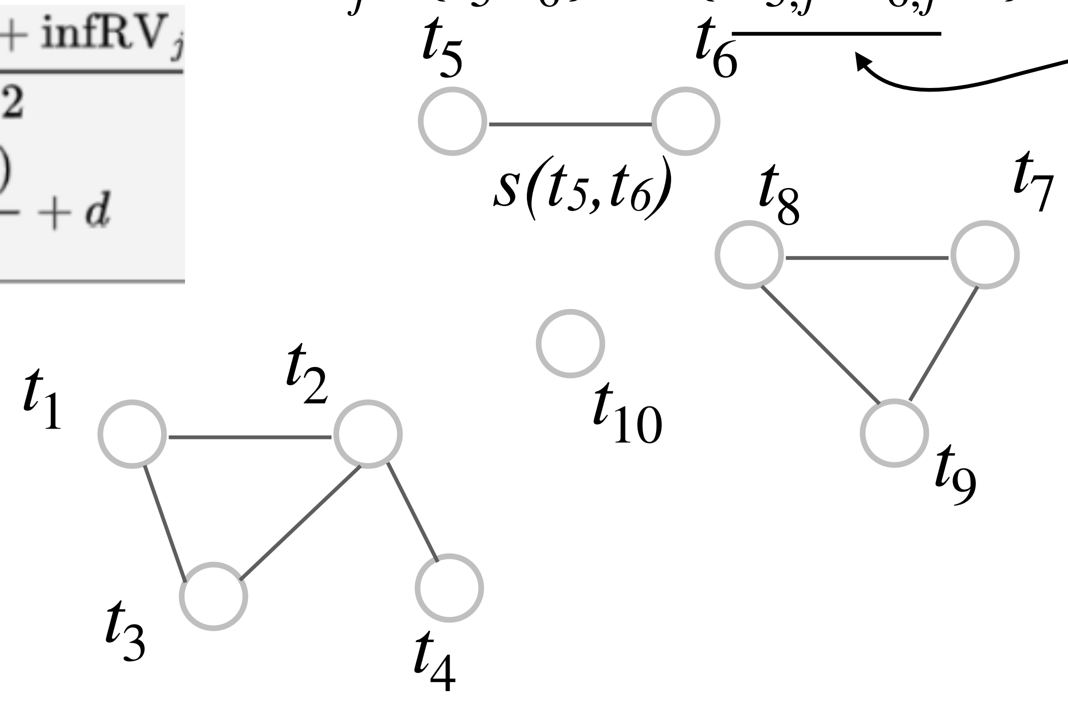
$$\mathscr{E}_j : \{t_5, t_6\} \rightarrow \{w_{5,j}, w_{6,j}; c\}$$

- Equivalence Class Count
- Conditional probabilities $w_{i,j}$ for transcript i in equivalence class j. It consolidates the probability of $\sum_{f_k \in \mathscr{E}_j} P\{f_k \mid t_i\}$

$$s(t_i, t_j) = \text{infRV}_{i+j} - \frac{\text{infRV}_i + \text{infRV}_j}{2}$$

$$\text{infRV}_i = \frac{\max(\sigma^2_{pi} - \mu_i, 0)}{\mu_i + pc} + d$$

The lower the edge weight, the better are those candidates for grouping.



Construct a graph G = (V, E) where :

V = set of transcripts

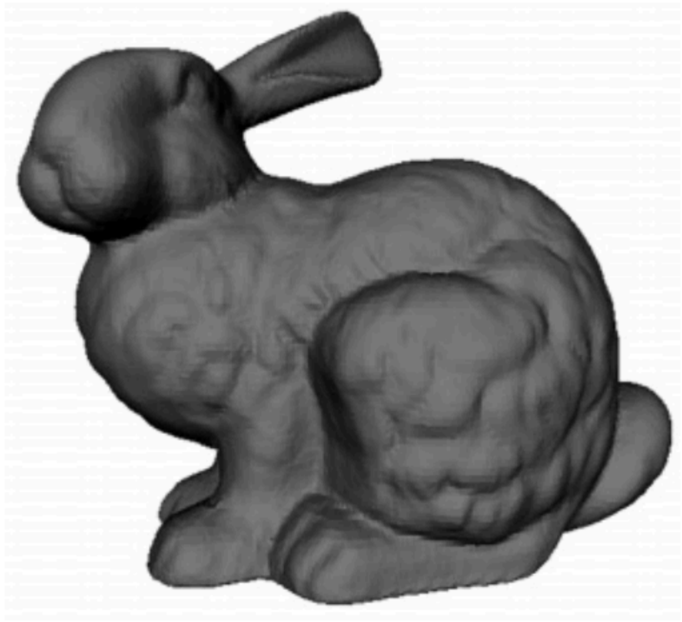
E = $\{v_i, v_j\}$ such that v_i, v_j co-occur within at least one equivalence class

Motivation: Transcripts sharing *no* reads are unlikely to be good candidates for merging

Discovering transcriptional groups is equivalent to contracting edges in this graph!

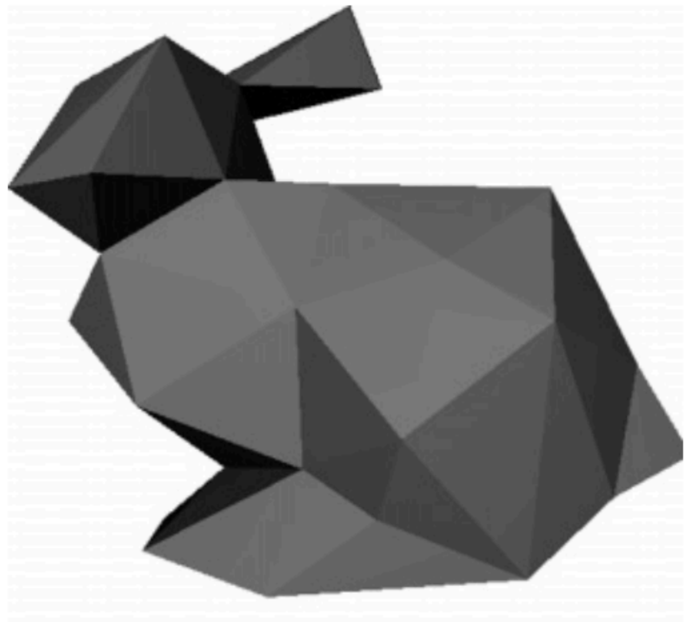
*This idea goes quite far back in the RNA-seq literature; at least to MMSeq (Turro et al. 2011)

Some interdisciplinary motivation



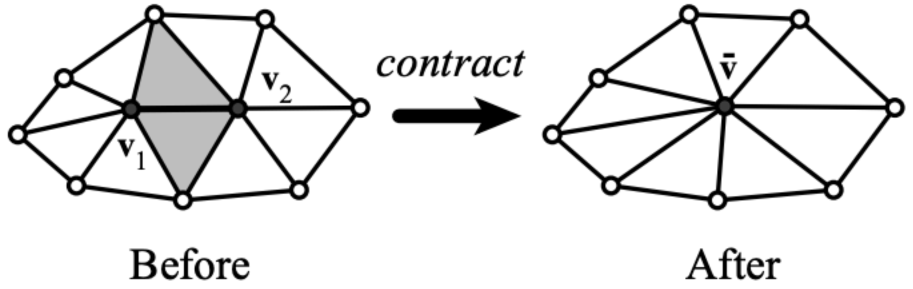
69,451 triangles to render original bunny

Approximation with 1000 triangles



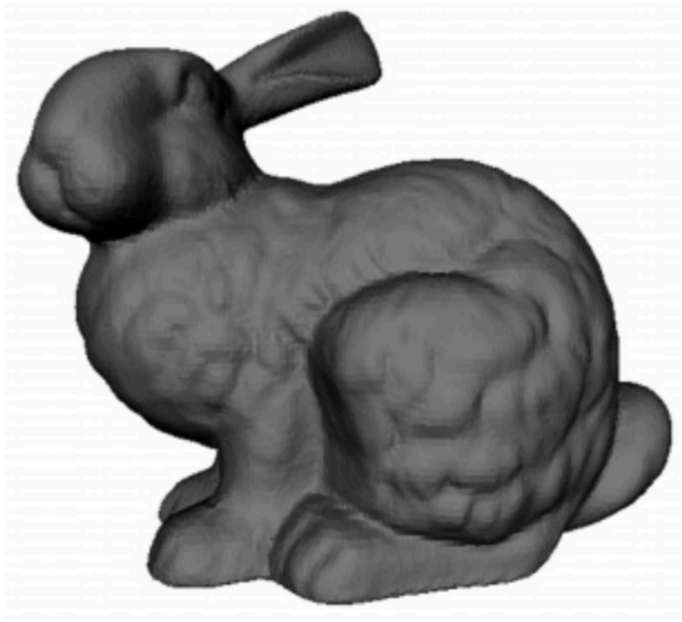
Approximation with 100 triangles

Goal : Simplify mesh by reducing # of vertices while maintaining visual fidelity.



Garland, Michael, and Paul S. Heckbert. "Surface simplification using quadric error metrics." *Proceedings of the 24th annual conference on Computer graphics and interactive techniques*. 1997.

Some interdisciplinary motivation



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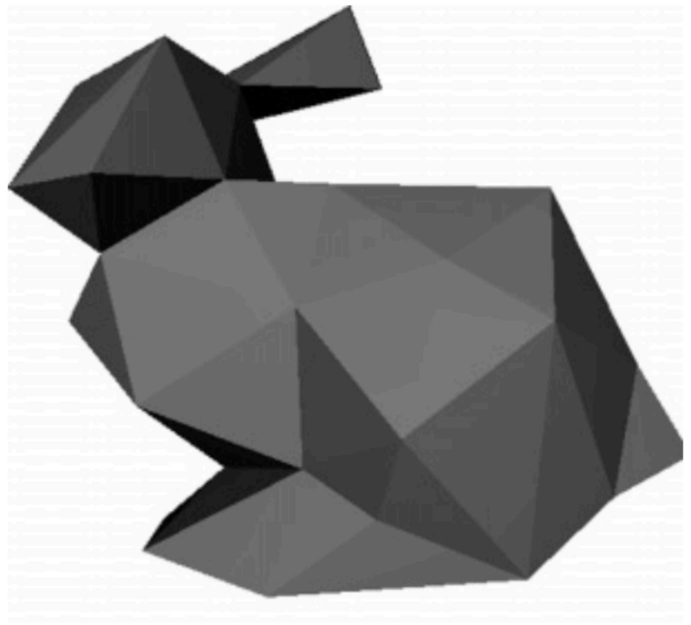
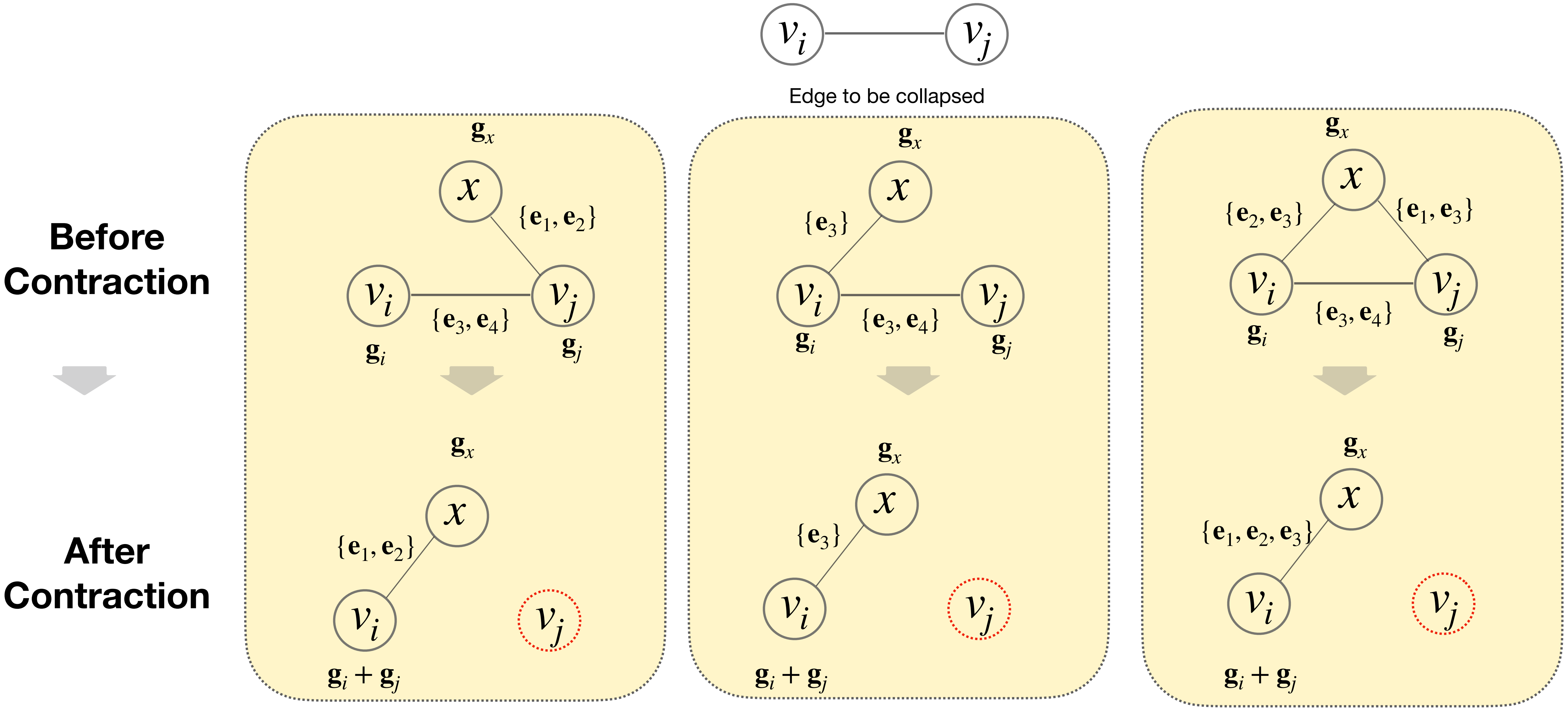


Approximation with 100 triangles

Approach : Put all edges in a min heap. While we have not exceeded our approximation budget, extract the edge whose collapse will make the *least* difference, contract the edge, evaluate the newly adjacent vertices, and place them back in the heap.

Garland, Michael, and Paul S. Heckbert. "Surface simplification using quadric error metrics." *Proceedings of the 24th annual conference on Computer graphics and interactive techniques*. 1997.

How to “contract” an edge

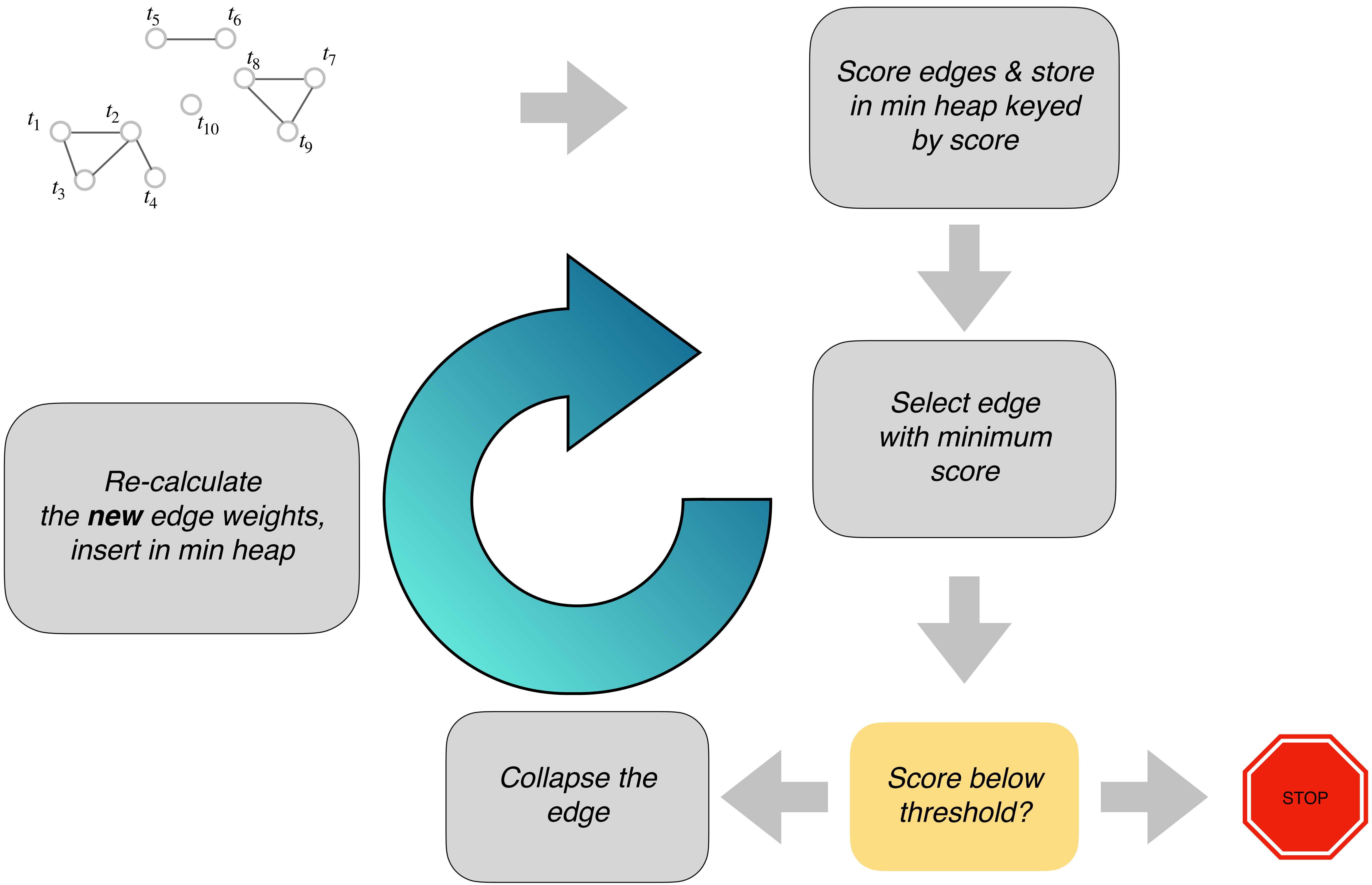


Case 1 *Case 2* *Case 3*

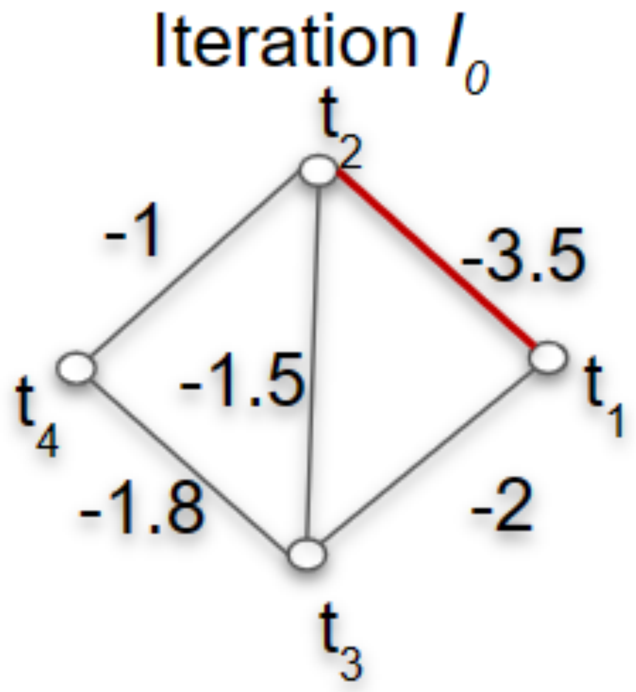
Updated Score

$$s(v_i, x) = \mathrm{infRV}(\mathbf{g_i} + \mathbf{g_j} + \mathbf{g_x}) - \frac{\mathrm{infRV}(\mathbf{g_i} + \mathbf{g_j}) + \mathrm{infRV}(\mathbf{g_x})}{2}$$

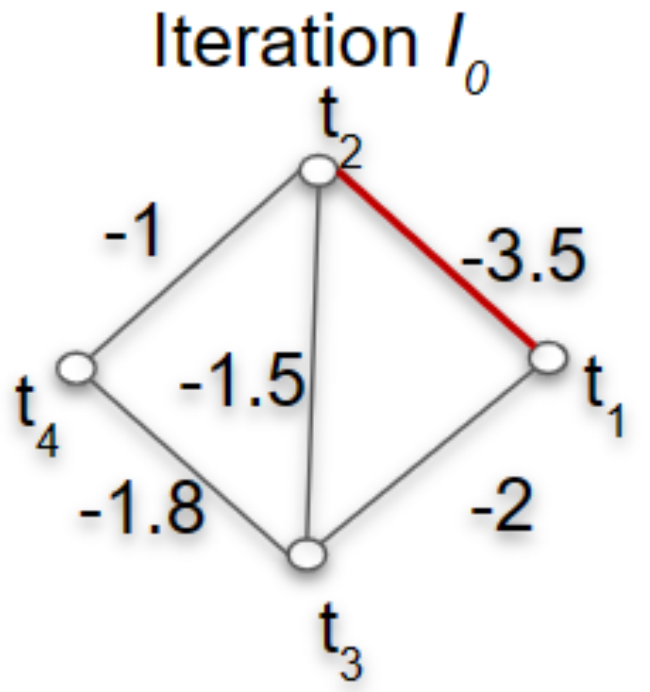
Overview of the algorithm



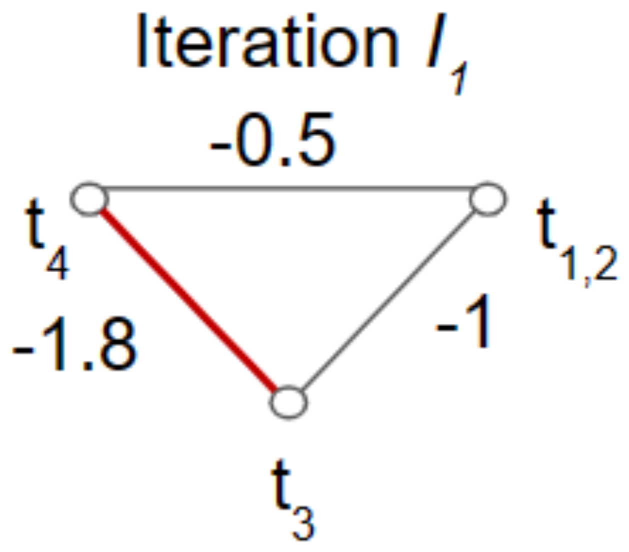
Example



Example

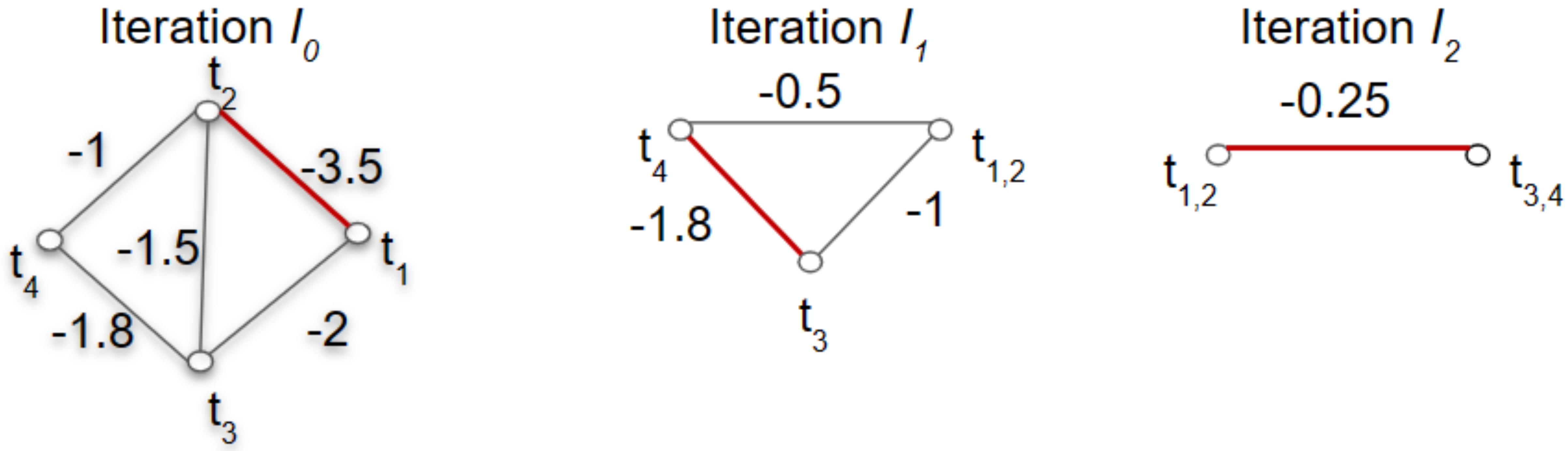


Iteration I_0
t_2
-1
-3.5
t_4
-1.5
t_1
-1.8
-2
t_3



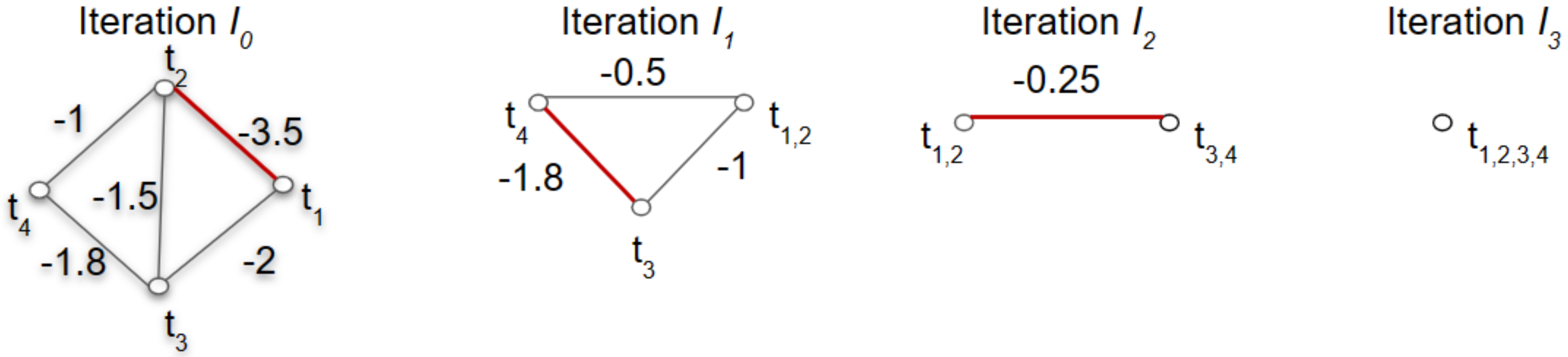
Iteration I_1
-0.5
t_4
$t_{1,2}$
-1.8
-1
t_3

Example



Iteration I_0
t_2
-1
-3.5
t_4
-1.5
t_1
-1.8
-2
t_3
Iteration I_1
-0.5
t_4
$t_{1,2}$
-1.8
-1
t_3
Iteration I_2
-0.25
$t_{1,2}$
$t_{3,4}$

Example

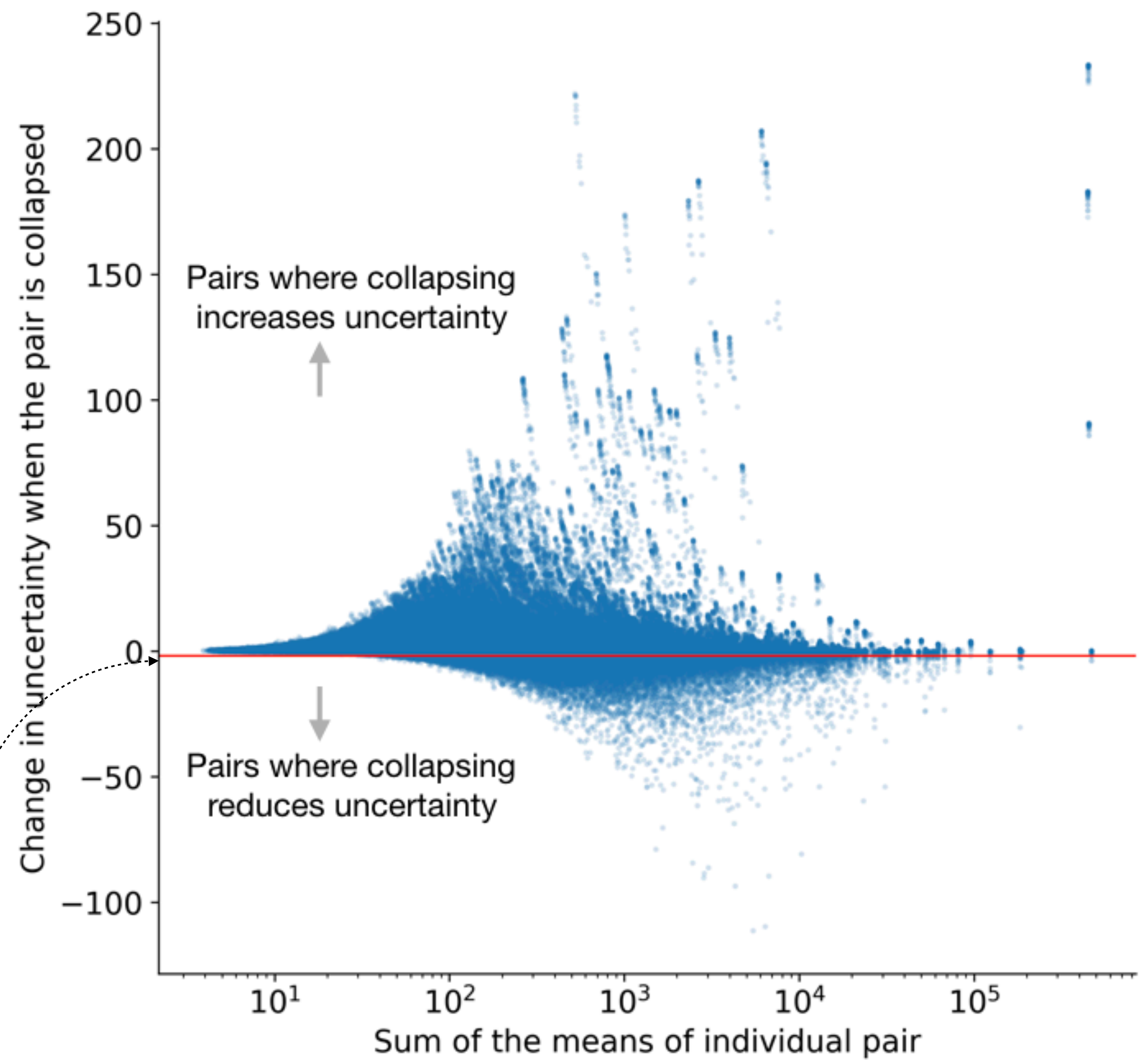


Iteration I_0
t_1
t_2
t_3
t_4
-1
-3.5
-1.5
-1.8
-2
Iteration I_1
-0.5
t_4
$t_{1,2}$
-1.8
-1
t_3
Iteration I_2
-0.25
$t_{1,2}$
$t_{3,4}$
Iteration I_3
$t_{1,2,3,4}$

Determining when to stop!

Data: $\mathcal{T}$, $\mathcal{G}$, selection_size
Result: τ_{new}
$\tau_{old} \leftarrow 0$;
$\tau_{new} \leftarrow 0$;
converged $\leftarrow$ False ;
$S \leftarrow \{(i,j) | i,j \in \mathcal{T}\}, s.t. |S| =$ selection_size ;
$\mathcal{B} \leftarrow \emptyset$;
while ***not*** converged **do**
    for $(i,j) \in S$ **do**
        $\mathcal{B} = \mathcal{B} \cup s(i,j)$;
    end
    $\tau_{new} \leftarrow$ percentile($\mathcal{B}$,2.5) ;
    diff $\leftarrow |\tau_{old} - \tau_{new}|$;
    if diff $< \delta$ **then**
        converged $\leftarrow$ True ;
    end
    selection_size $\leftarrow$ selection_size $\cdot 2$;
    $\tau_{old} \leftarrow \tau_{new}$;
    $\mathcal{B} \leftarrow \emptyset$;
end
return τ_{new} ;

Algorithm 1: Threshold τ selection algorithm



Determining when to stop!

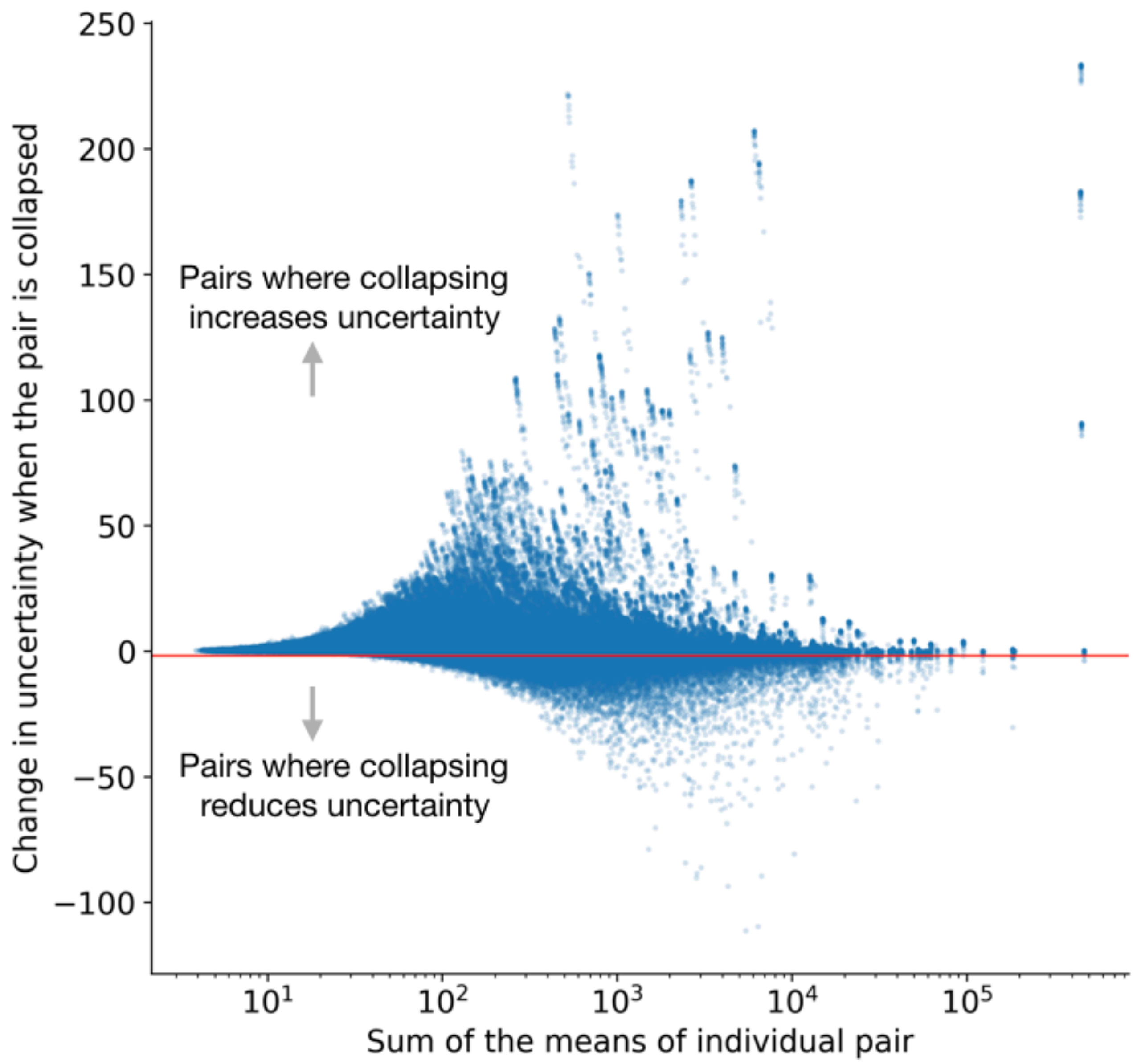
Basically:

Define the background distribution as the distribution of “scores” for random transcript pairs.

Estimate the tail (2.5%) of this distribution.

Repeatedly double the size of the background distribution until this quantile stops changing.

This is our threshold.



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  if diff $< \delta$ **then**
    converged ← True ;
  end
  selection_size ← selection_size ·2 ;
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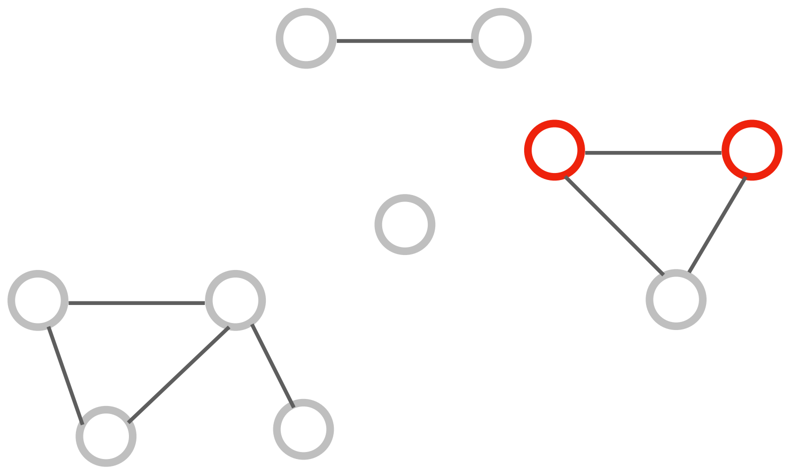
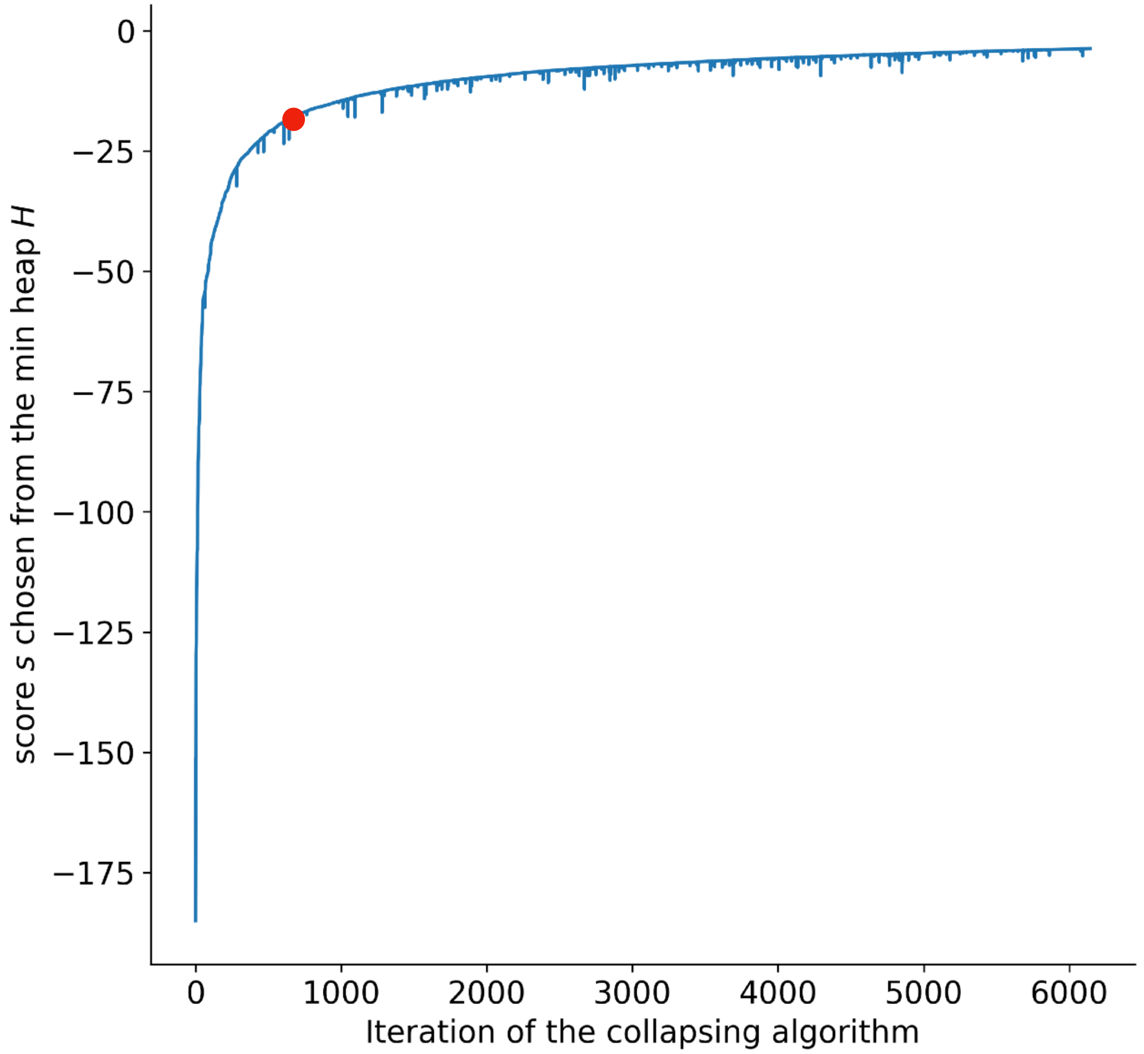
end
return τ_{new} ;

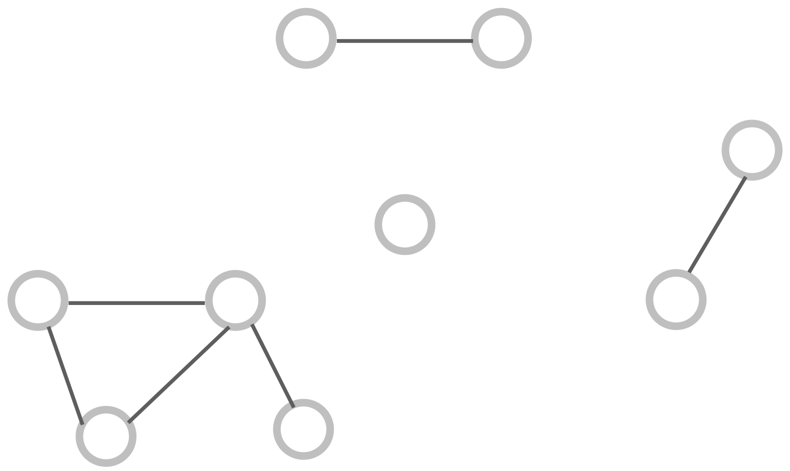
Algorithm 1: Threshold τ selection algorithm

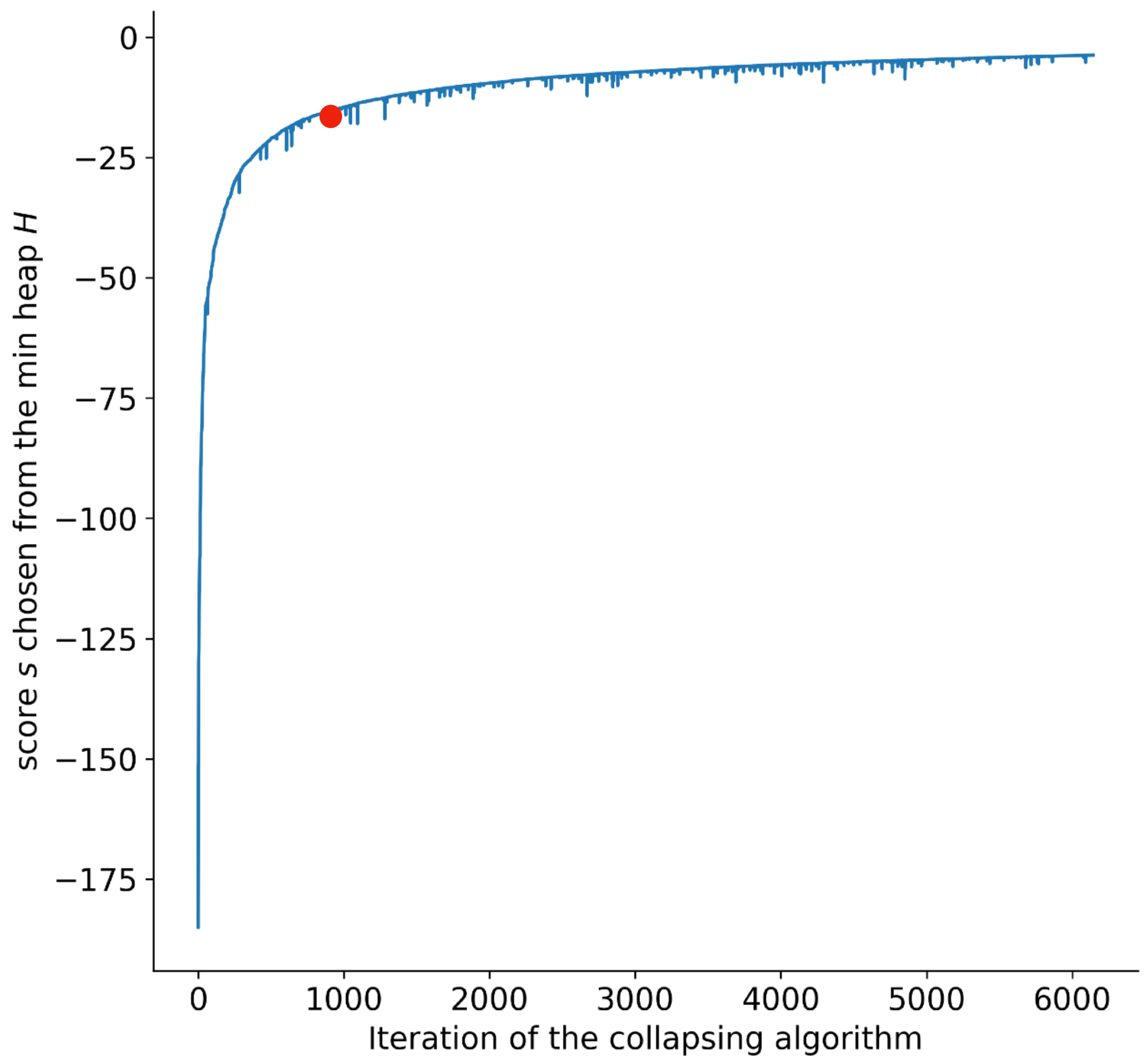
Progress over iterations



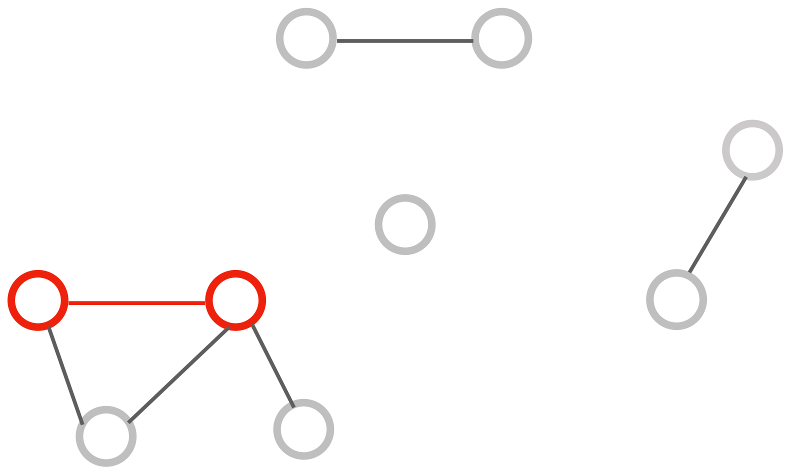


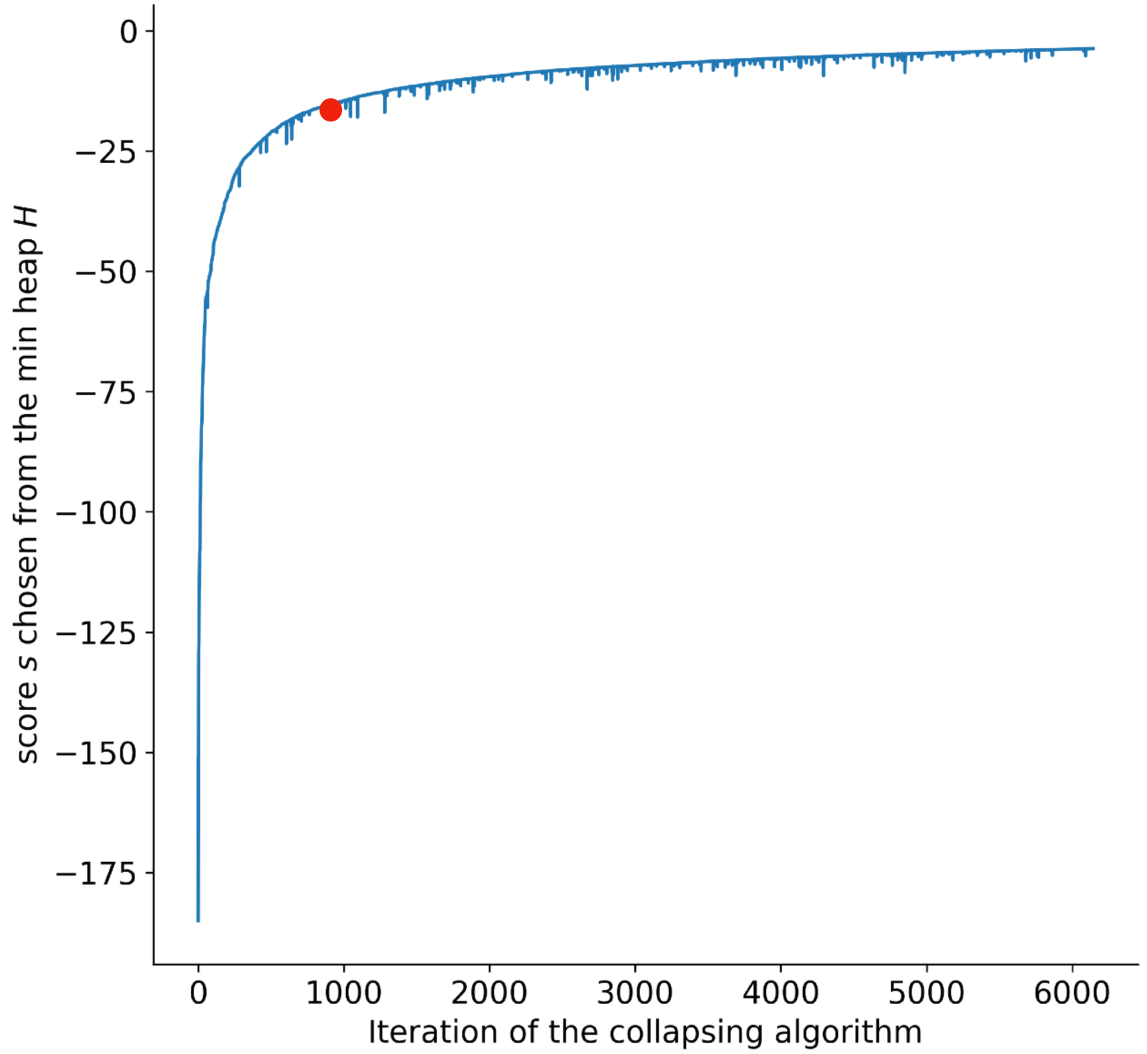
Progress over iterations





Progress over iterations





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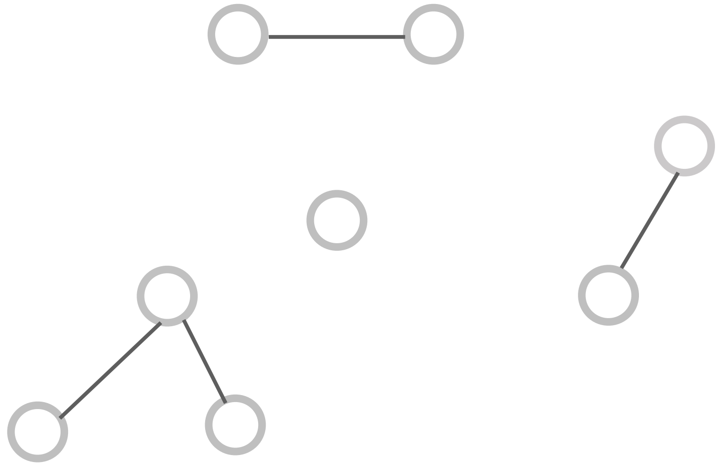
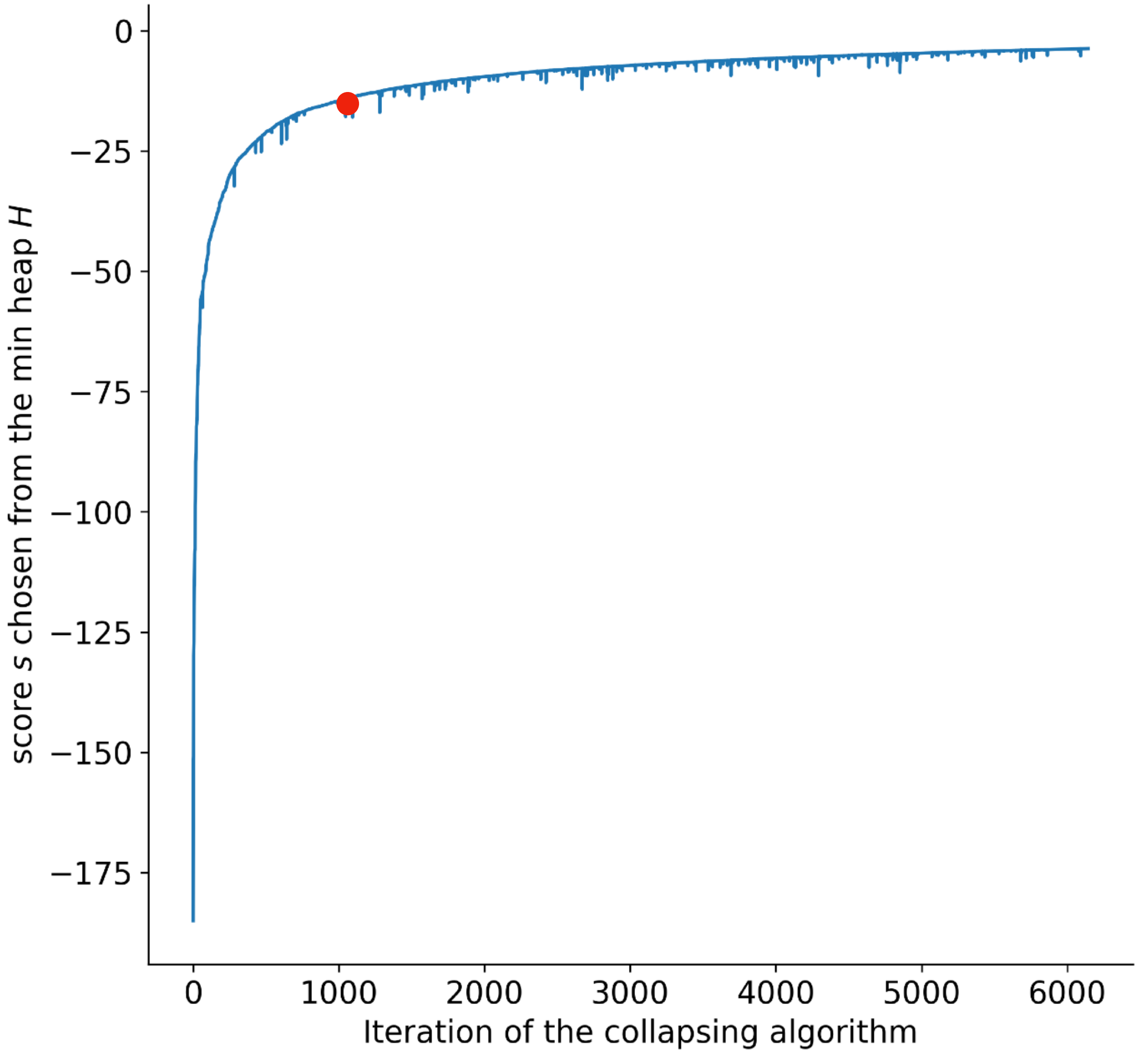




Terminus for multiple samples

So far the algorithm we have presented, works for only a single RNA-Seq sample. This step is called for a group step. The output is a set of discrete transcript groups.

However, an RNA-Seq experiment consists of multiple samples. We want such discrete transcript groups for an experiment across multiple samples. Terminus provides a **consensus** step.

Consensus

The consensus procedure starts with individual groups, and constructs a union graph by treating each of the groups as a complete connected undirected graph (clique). For example: given two different groups $\{v_i, v_j, v_k\}$ from one sample and $\{v_i, v_j\}$ from another, terminus first constructs a weighted triangle with end points v_i, v_j and v_k where each edge has a weight 1. While considering the second group, terminus increases the weight of edge $\{v_i, v_j\}$ by 1. This iterative procedure generates a weighted graph ensuring any edge in the graph represents two transcripts that belong to the same group in at least one sample. Terminus further prunes the union graph and removes the edges that have a weight below a user-defined threshold. The consensus mechanism ensures that a pair of transcripts should at least co-occur in a specified number of samples to qualify for the final grouping.

How do terminus groups compare to individual transcripts?

To evaluate we looked at simulated & experimental data

- full-transcriptome simulated data in Human

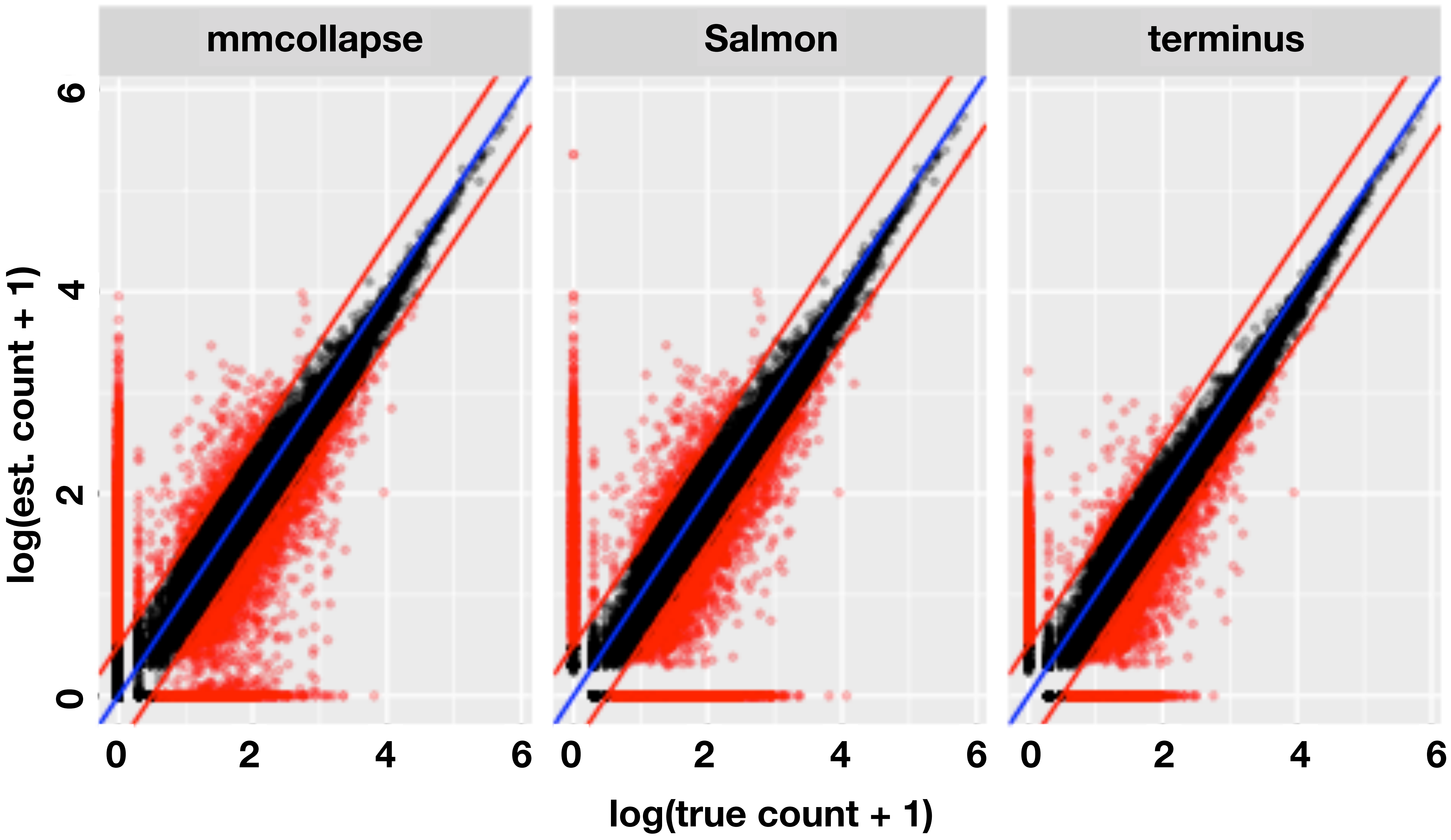
  multi-sample 4x4

- experimental data in Drosophila

  subset of “pasilla”* dataset

*Brooks, A. N. *et al*. (2011). Conservation of an RNA regulatory map between drosophila and mammals. *Genome Research*, 21(2), 193–202.

How do terminus groups compare to individual transcripts?



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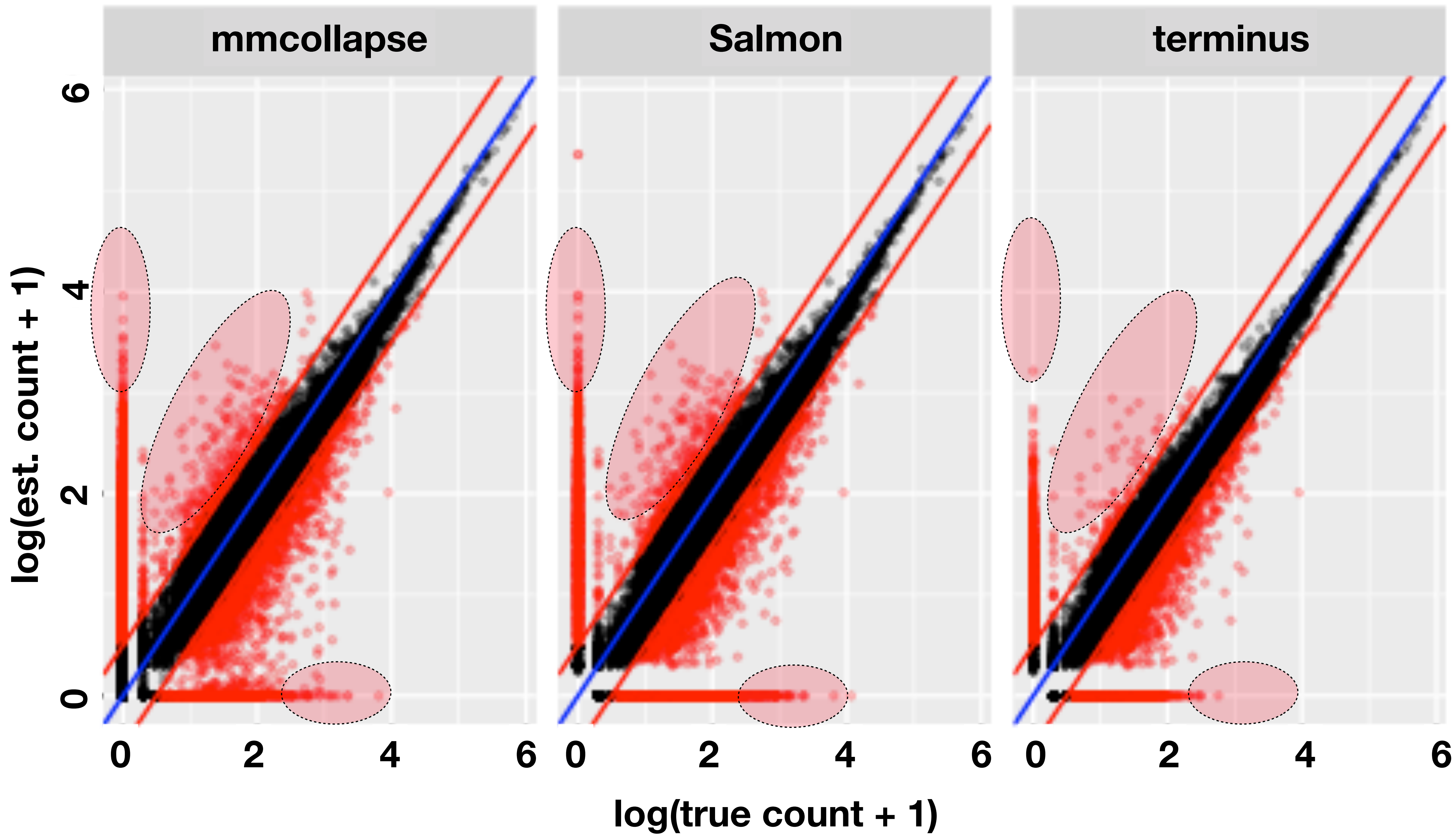

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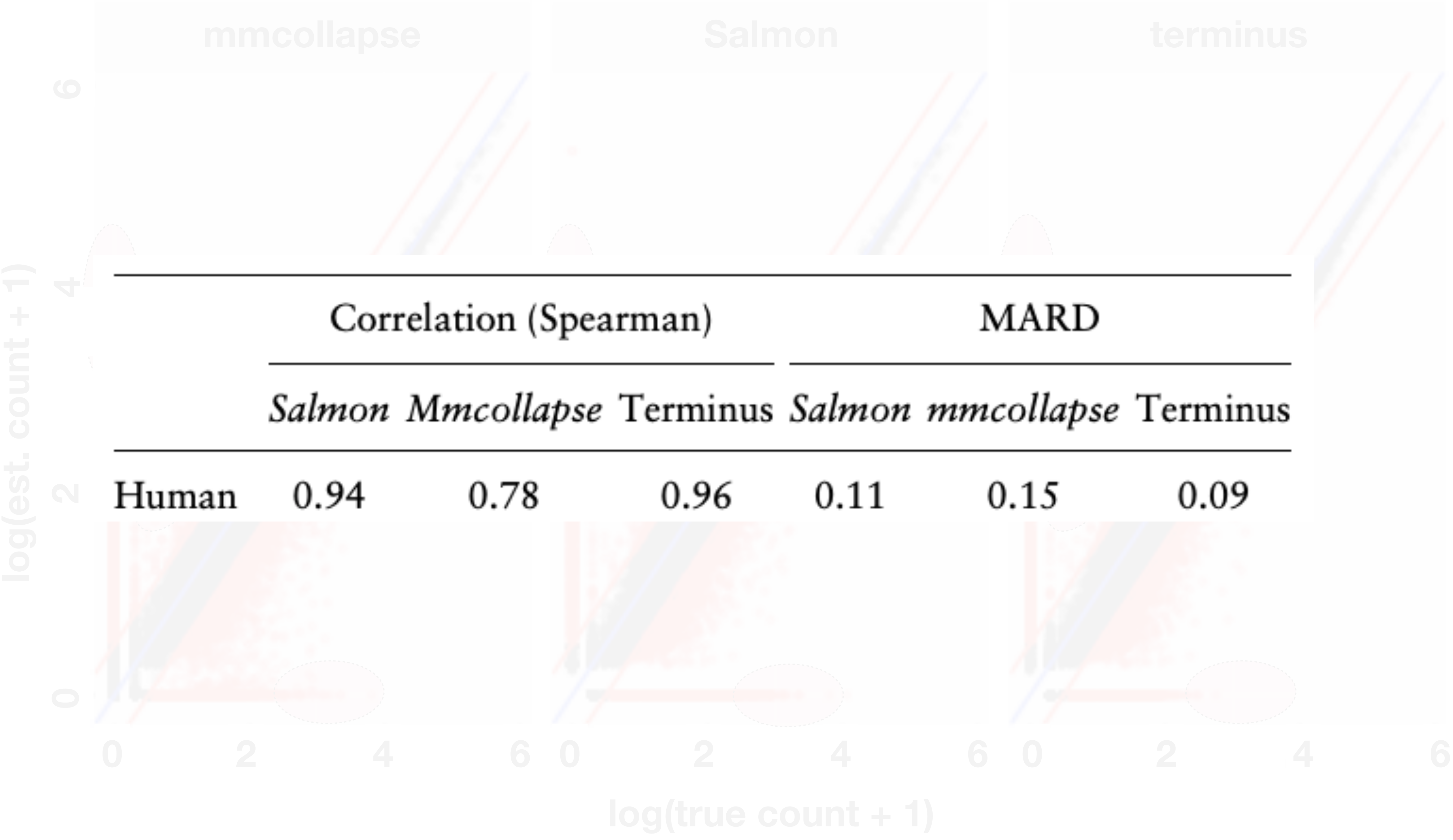
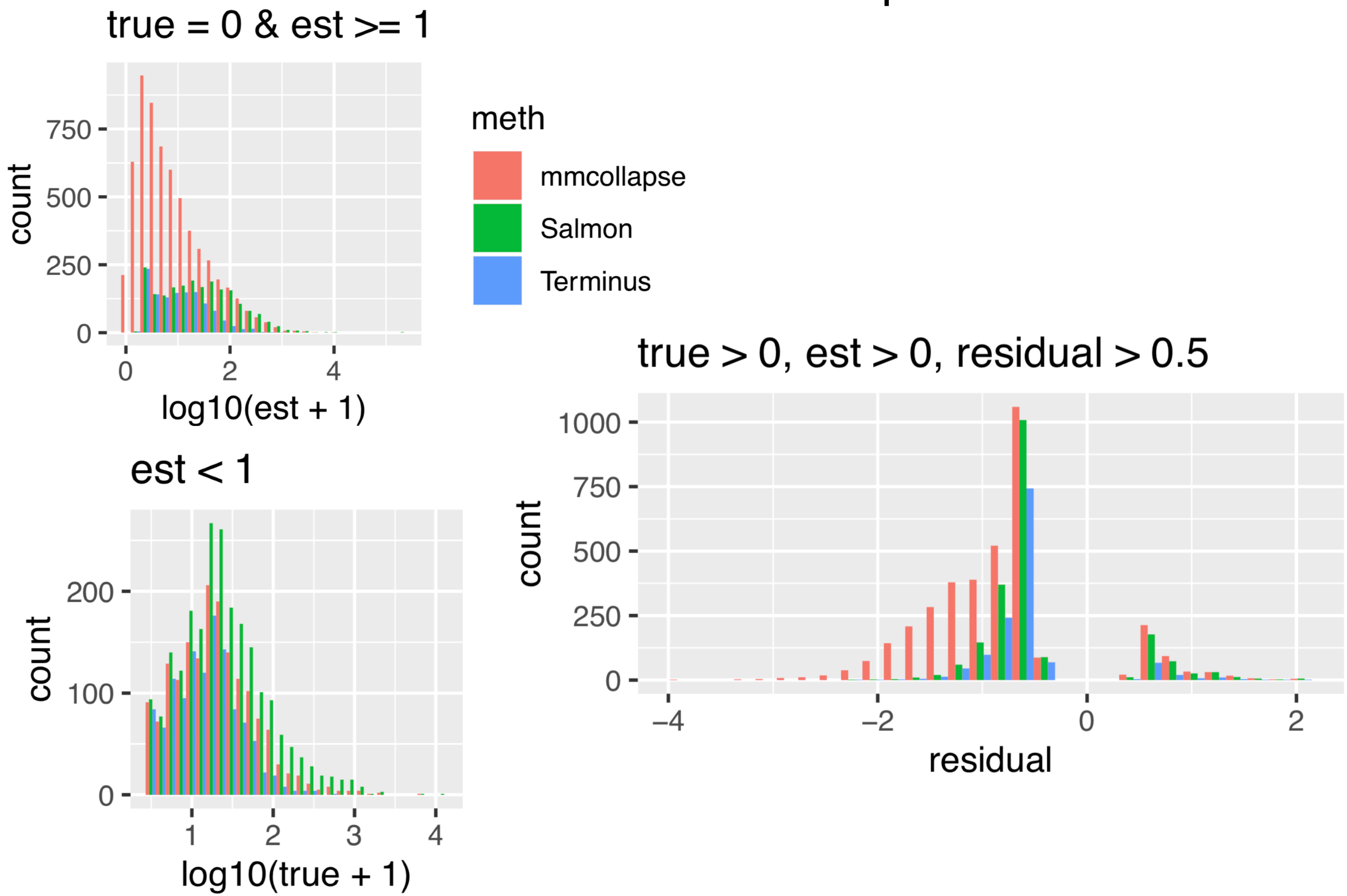


	Correlation (Spearman)			MARD		
	Salmon	*Mmcollapse*	Terminus	*Salmon*	*mmcollapse*	Terminus
Human	0.94	0.78	0.96	0.11	0.15	0.09

How do terminus groups compare to individual transcripts?



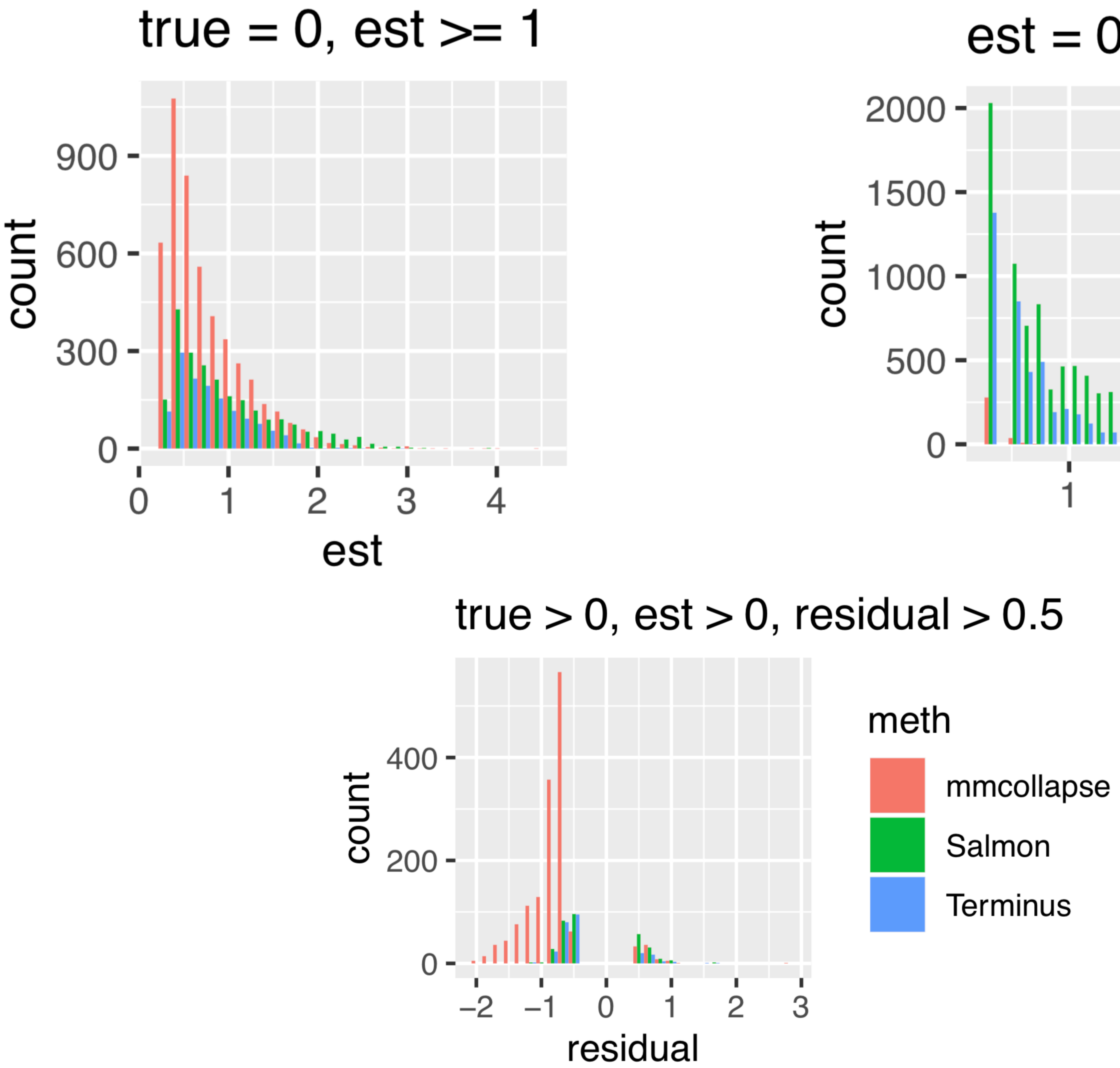
The effect of group when estimating allele-specific expression

Allele-specific expression is *especially* challenging, since the transcripts corresponding to the alleles of a gene are often *nearly identical* (if not actually identical). This exacerbates problems associated with uncertainty!

Used the reference mouse transcriptome, annotated variants, and g2gtools* to build a diploid NOD/PWK reference. Quantify transcript abundances *including* all allelic variants.

https://github.com/churchill-lab/g2gtools

Allele-specific expression



The effect of group when estimating allele-specific expression

Define an *incorrect dominant allele* as a situation where in estimation:

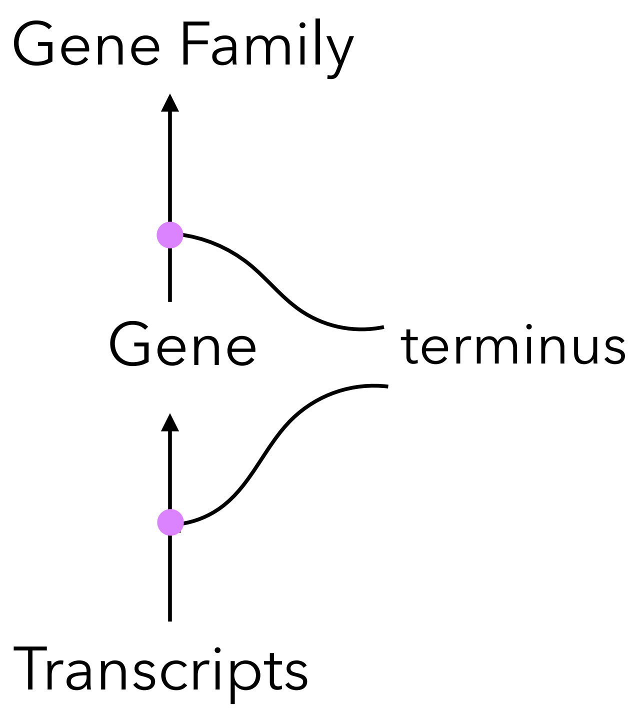
est(allele A) > est(allele B) while true(allele B) > true(allele A)

In other words, these are dominant allele switches.

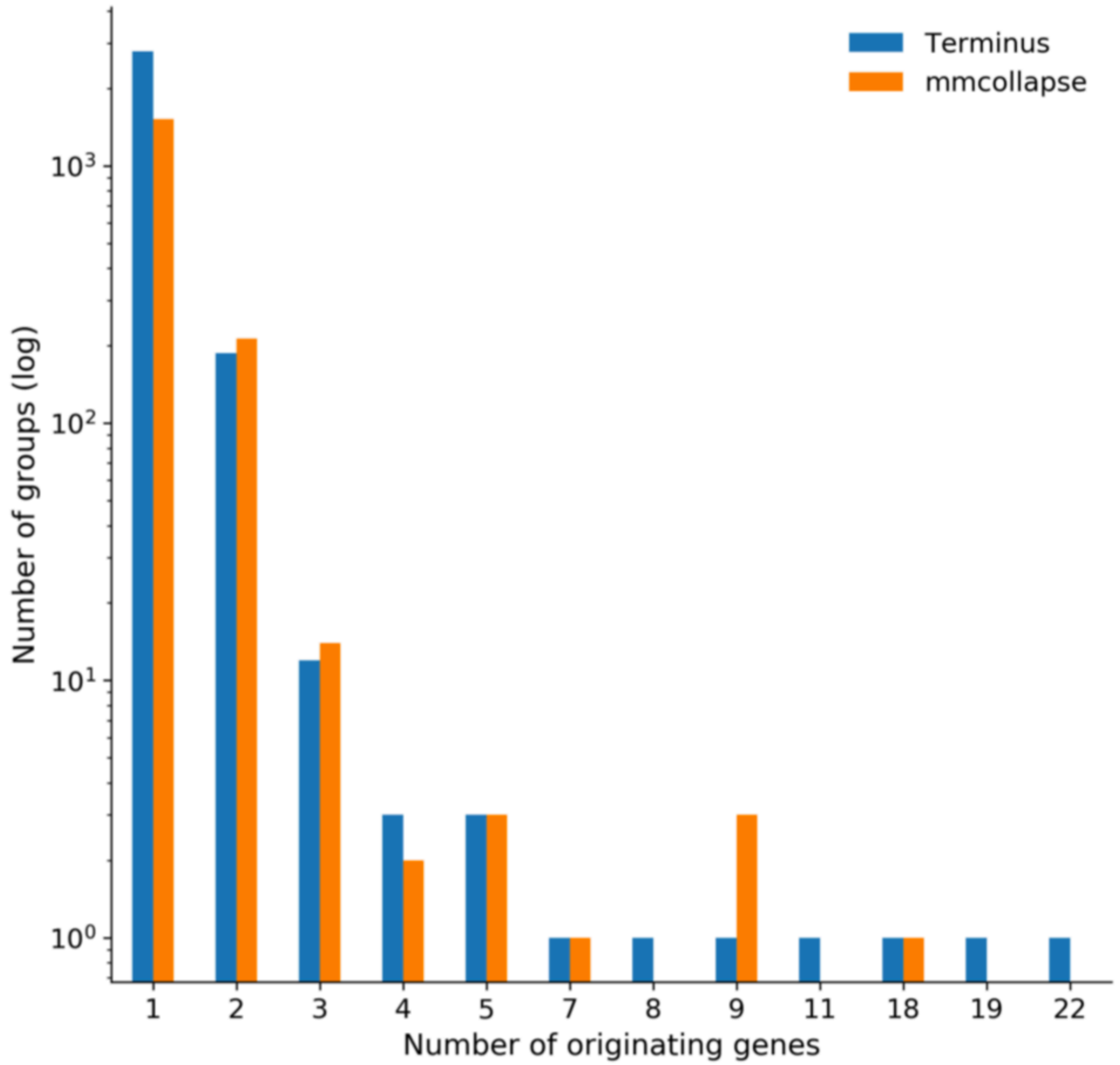
	Salmon	mmcollapse	terminus
Fraction of wrong dominant allele	0.1	0.04	0.04

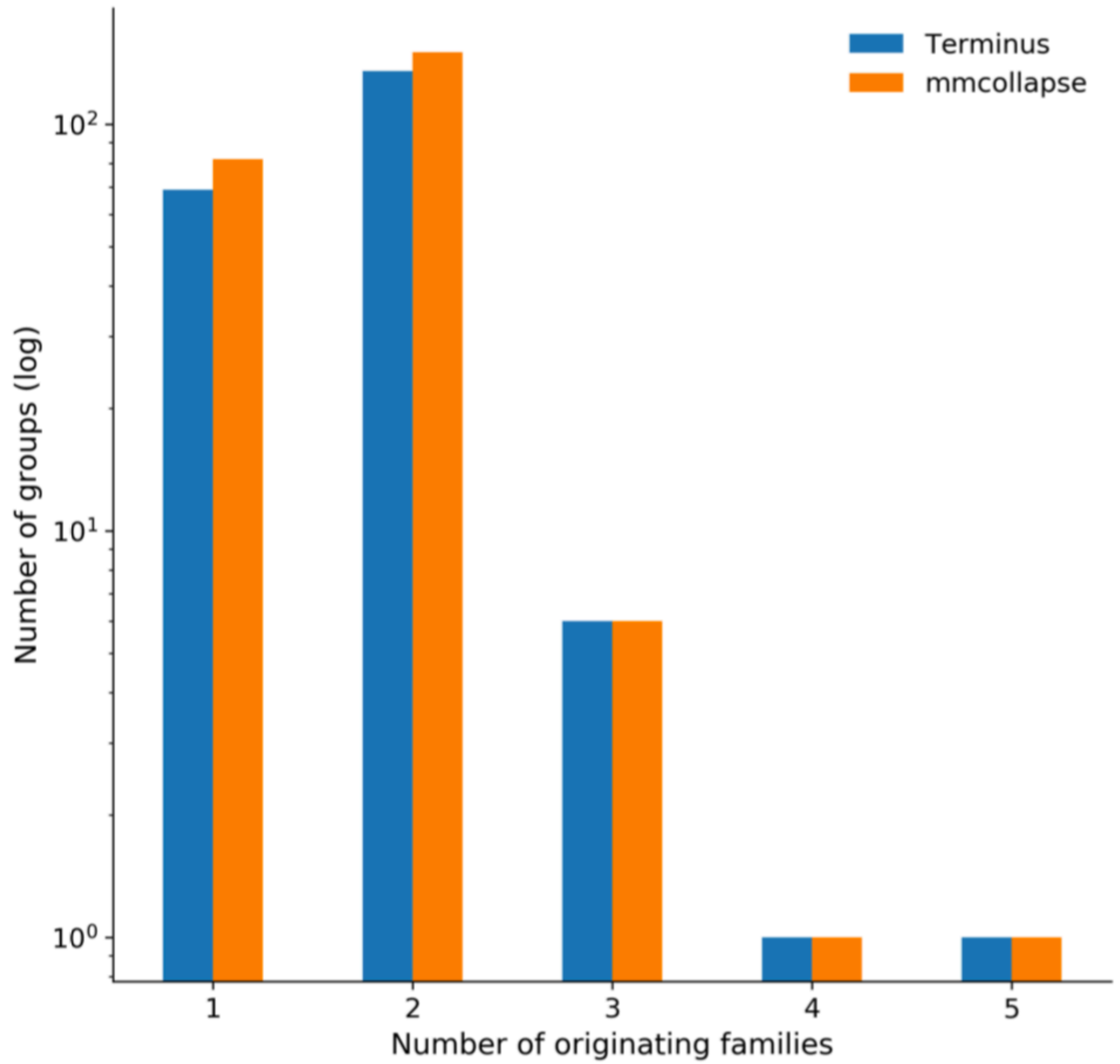
Both mmcollapse & terminus have fewer than *half* as many wrong dominant alleles.

Biological plausibility of groups on experimental data



In experimental data, we have no ground truth. But, we can look at how “reasonable" the groups seem.





- Terminus groups are *data-driven*.
- They cannot be replaced via any annotation.
- Though they can span genes, they most often are from one gene or gene family.

Biological plausibility of groups on experimental data

In experimental data, we have no ground truth. But, we can look at how “reasonable" the groups seem.

The groups are most-frequently transcripts of a single gene. When they are not, they usually come from 1 or 2 closely-related gene families.

e.g. One giant group had 54 transcripts! They turned out to all be from the “para” gene, which is annotated with 60 (highly-similar) transcript isoforms*!

*https://useast.ensembl.org/Drosophila_melanogaster/Gene/Summary?db=core;g=FBgn0285944;r=X:16455027-16533096

Terminus is fast & memory efficient!

Datasets	Memory(MB)		Time (h:m:s)	
	mmcollapse	Terminus	mmcollapse	Terminus
Human	370,675	2,841	3:46:37	2:04
Mouse	225,457	485	38:55:44	0:12
Drosophila	4,104	310	8:10	0:10

These time and memory profiles are *just* to perform the grouping.

As the annotation complexity and number of expressed transcripts increases, the resource requirements of mmcollapse become untenable. This is a result of the *algorithm*, not the implementation.

Can we have the best of both the worlds?

Transcript groups as a new entity/feature for analysis.

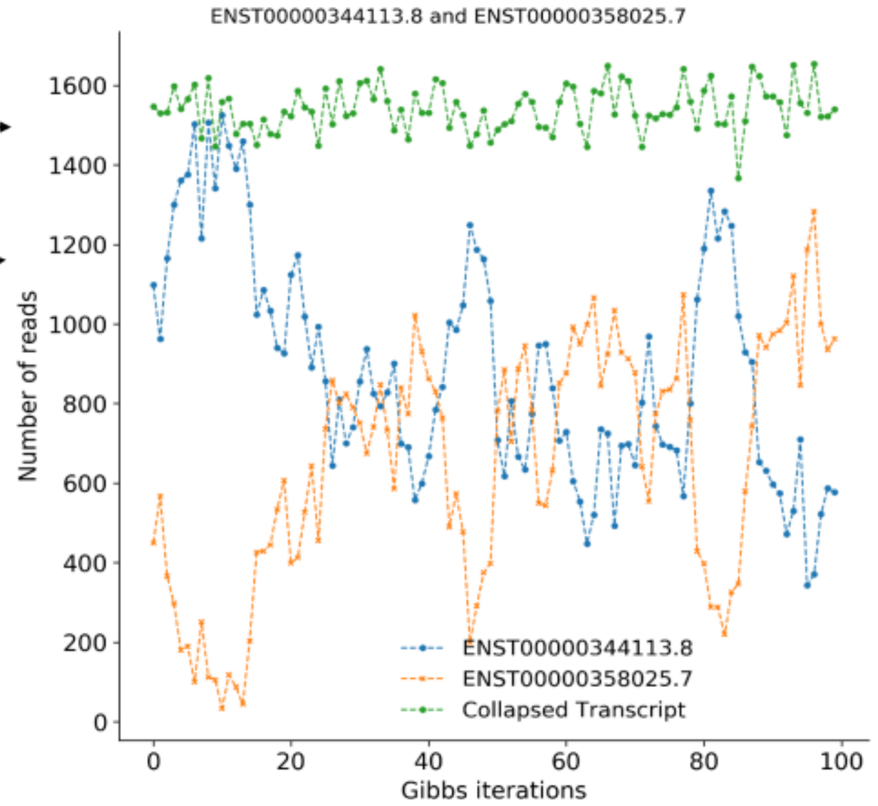
Flexible analysis of RNA-seq data using mixed effects models FREE
Ernest Turro, William J. Astle, Simon Tavaré Author Notes
Bioinformatics, Volume 30, Issue 2, January 2014, Pages 180–188,

Terminus enables the discovery of data-driven, robust transcript groups from RNA-seq data
Hirak Sarkar, Avi Srivastava, Héctor Corrada Bravo, Michael I Love, Rob Patro
Bioinformatics, Volume 36, Issue Supplement_1, July 2020, Pages i102–i110,

Terminus/mmcollapse proposed to **aggregate** the highly uncertain transcripts together into **groups** such that we are more certain about the abundance estimates of the transcript group**.**

Actual Gibbs traces for a pair of transcripts merged by Terminus.

Actual Gibbs traces for a pair of transcripts merged by our algorithm.



All TreeTerminus slides courtesy of Noor Pratap Singh

TreeTerminus

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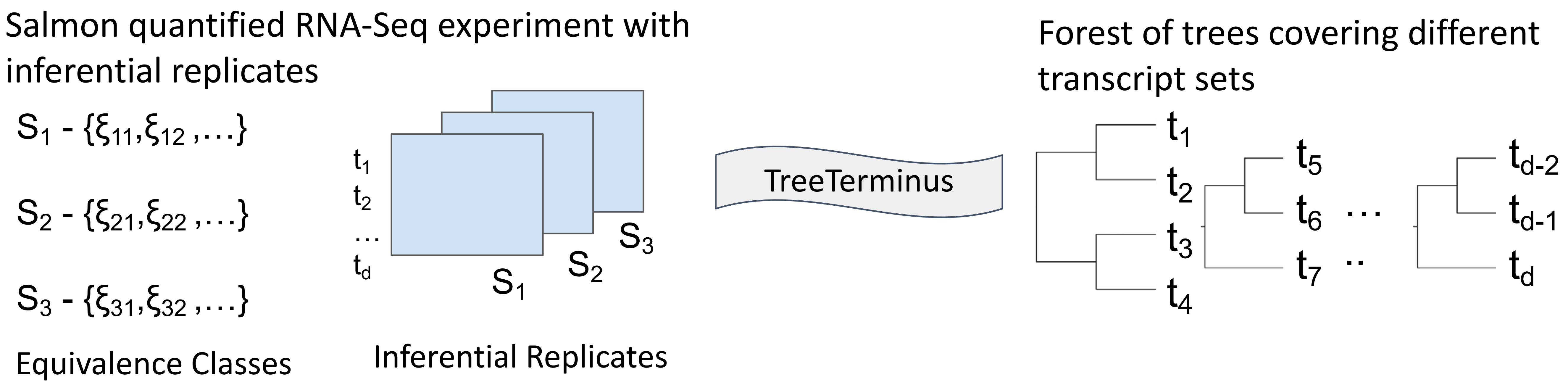

Patro, Rob, et al. "Salmon provides fast and bias-aware quantification of transcript expression." *Nature methods* 14.4 (2017): 417-419.

TreeTerminus

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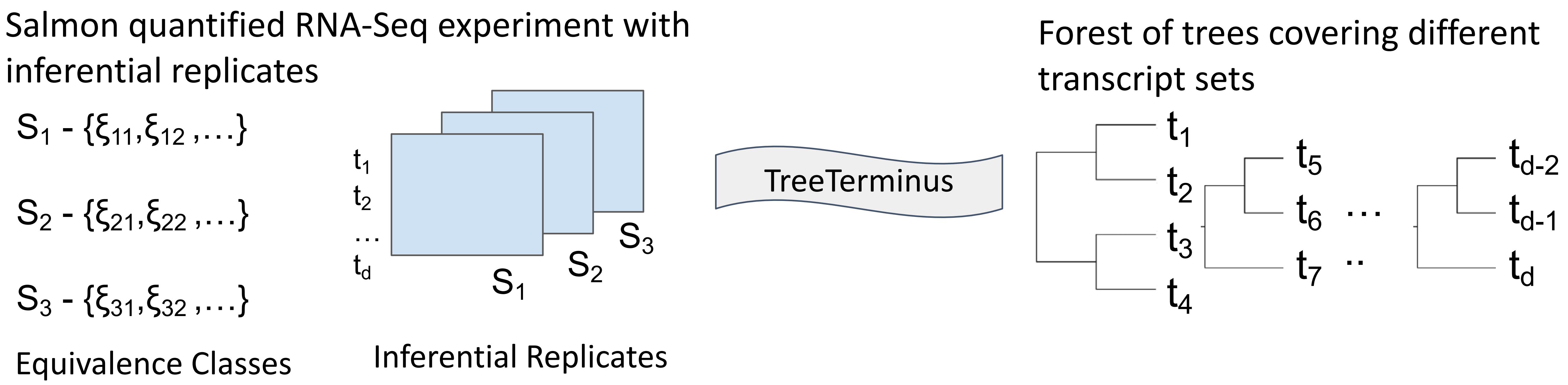

Leaves represent individual transcripts and internal nodes represent an aggregated set of leaf nodes.

Trees encode the whole hierarchy in which transcripts should be aggregated, such that the infRV decreases on an average as we ascend the tree topology.

Patro, Rob, et al. "Salmon provides fast and bias-aware quantification of transcript expression." *Nature methods* 14.4 (2017): 417-419.

Why care about the tree structure?

Dependent on a threshold.

Not enough flexibility with the end user.

Discrete groups are not always optimal for downstream tasks.

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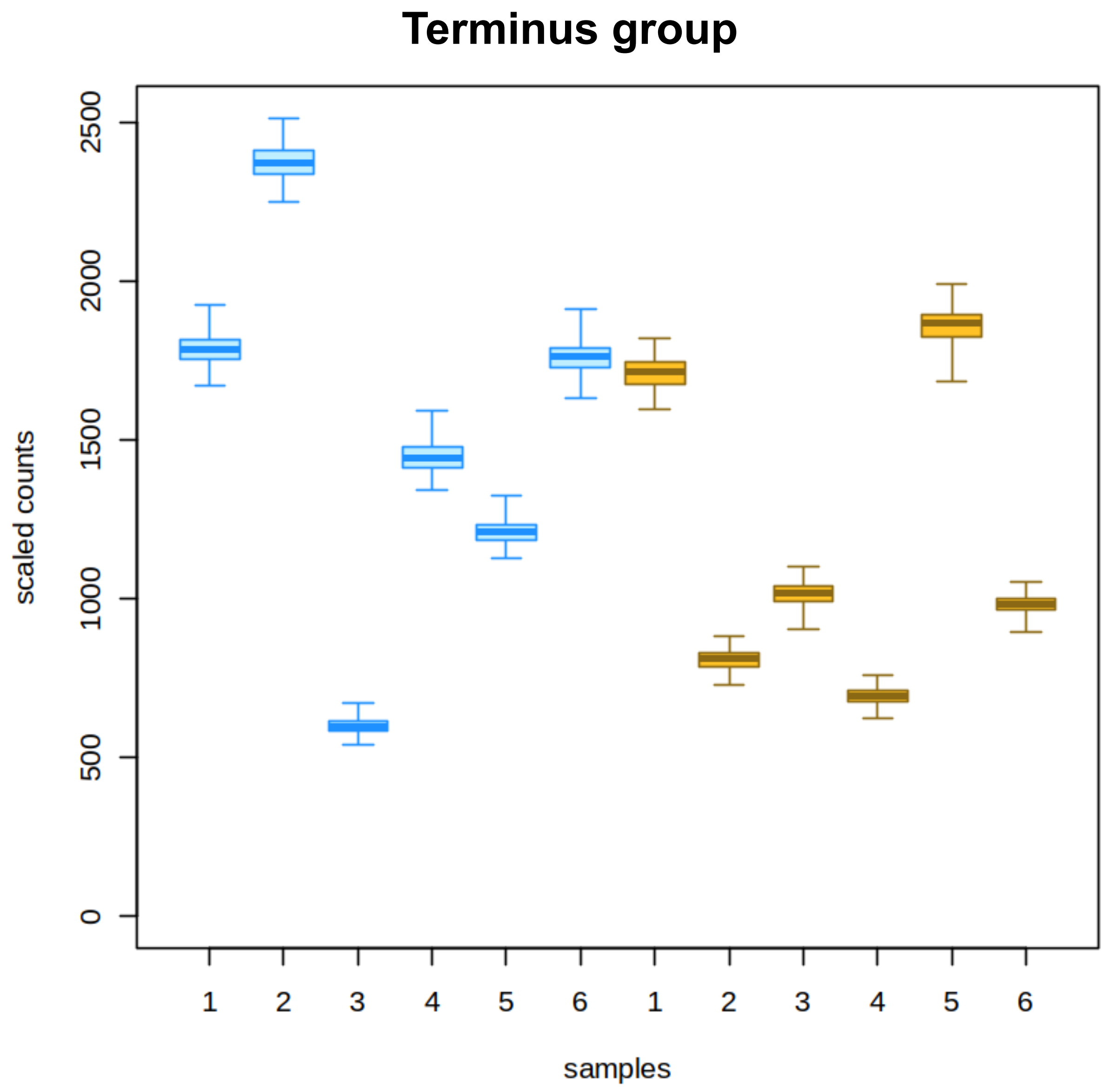

('ENST00000271751.9', 'ENST00000639952.1', 'ENST00000625140
lfc = -0.49, qvalue = 0.71

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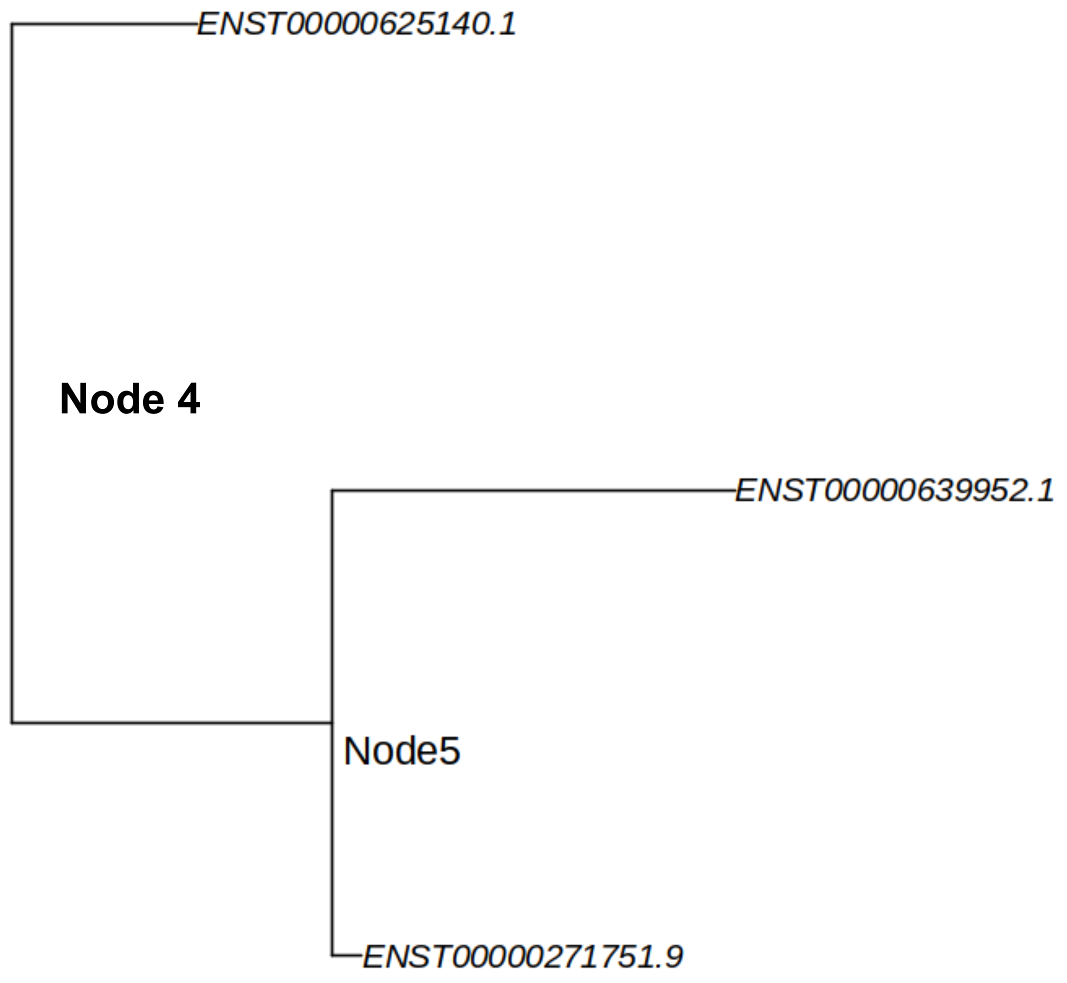

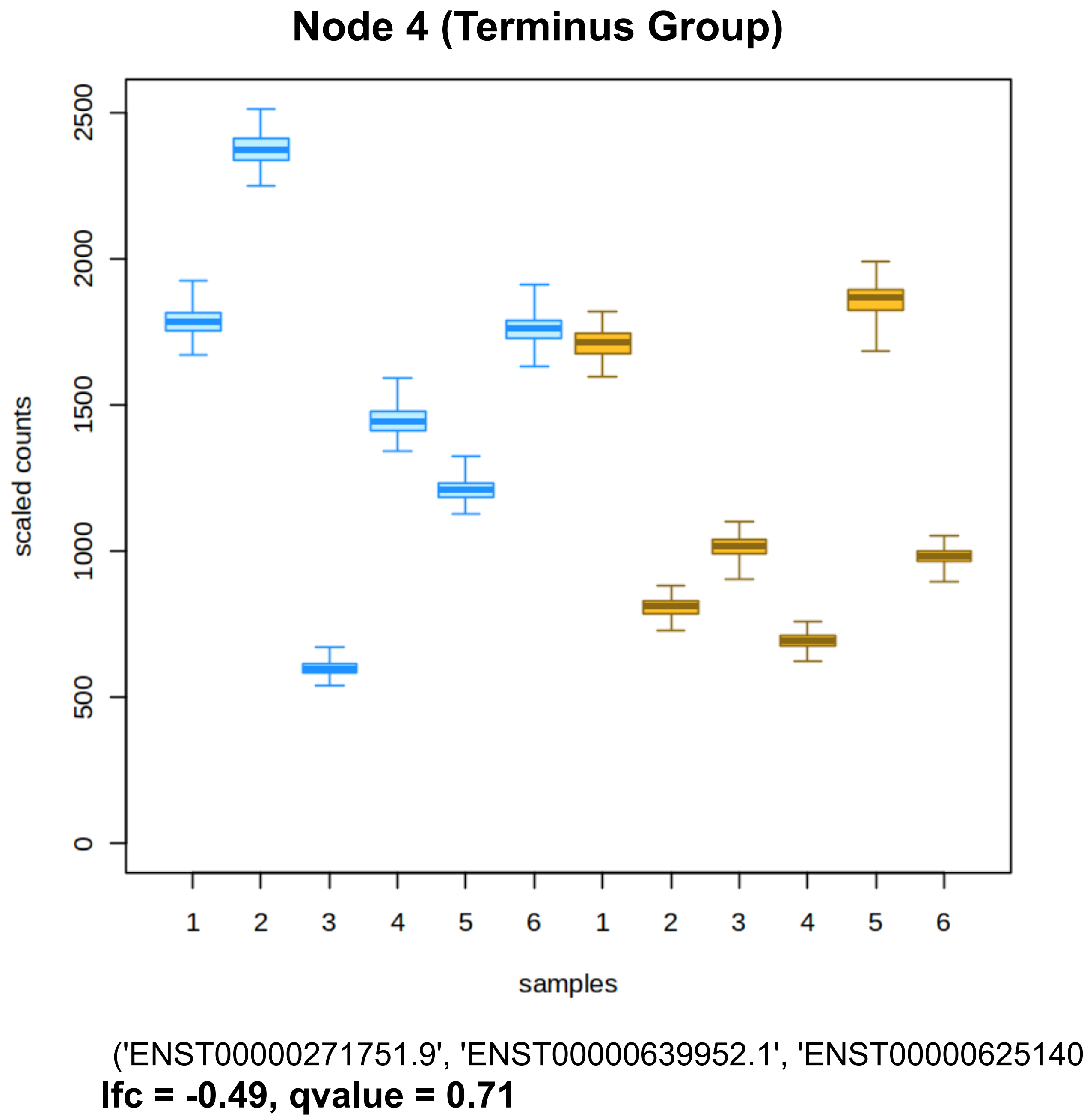

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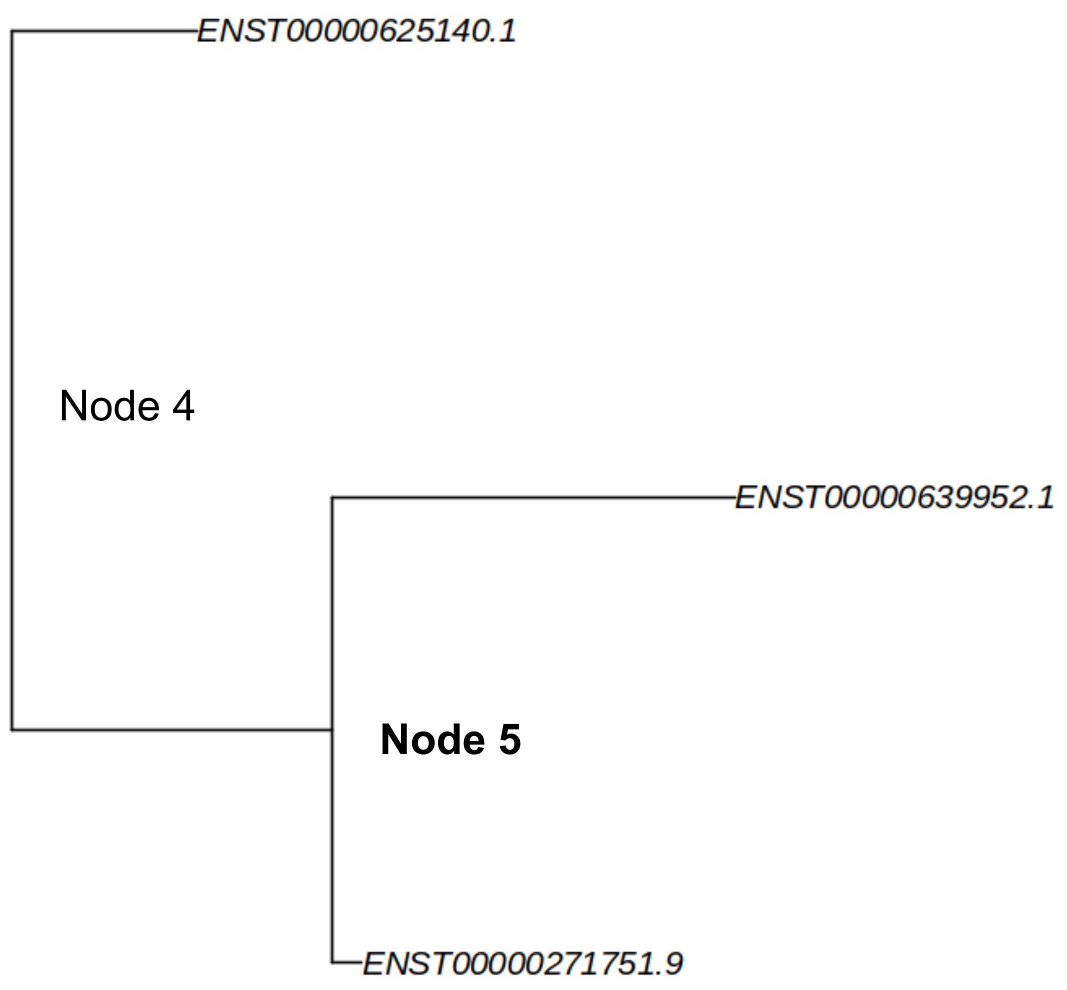

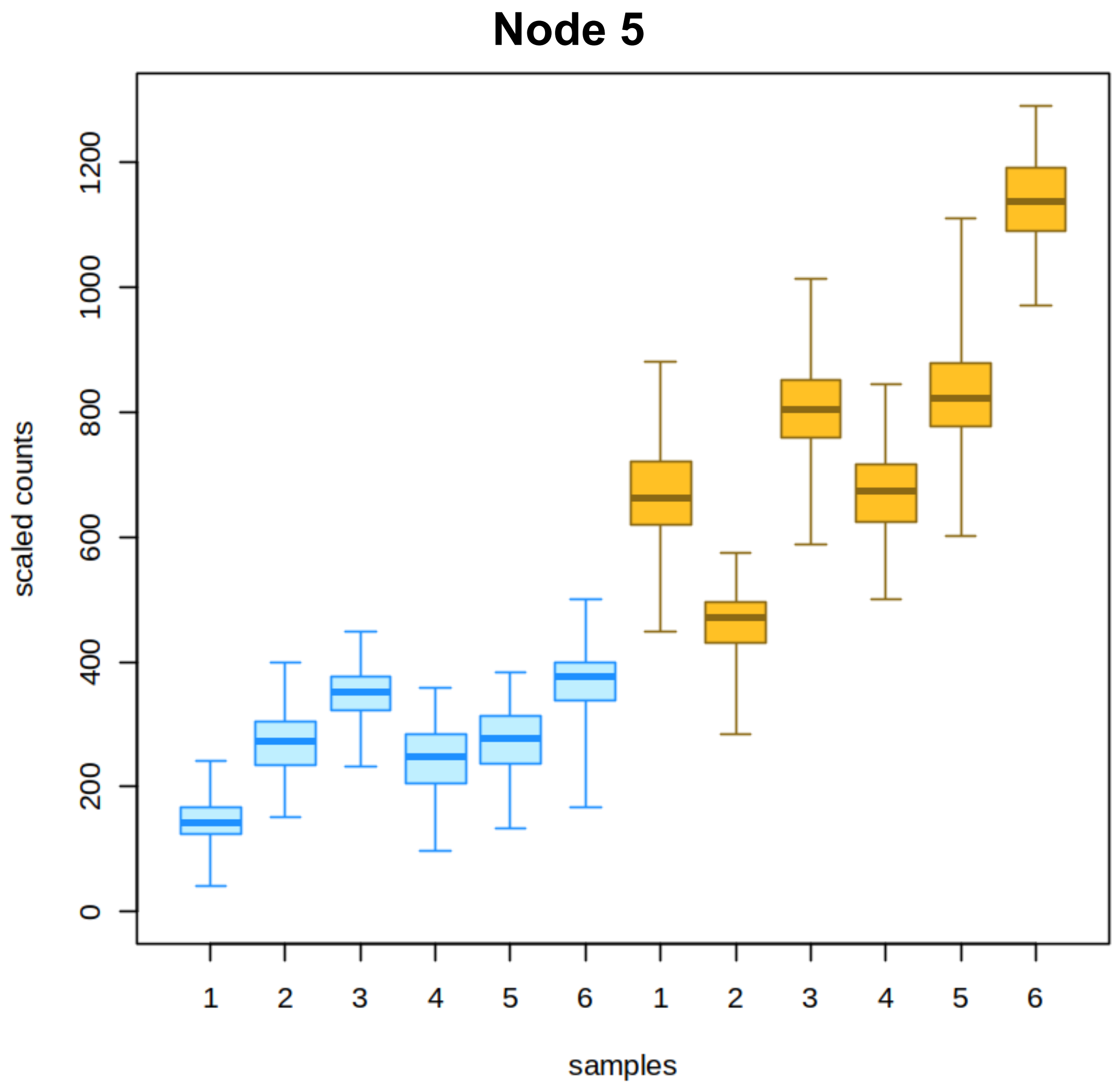

(('ENST00000271751.9', 'ENST00000639952.1',)

lfc = 1.43, qvalue = 0.007

Why care about the tree structure?

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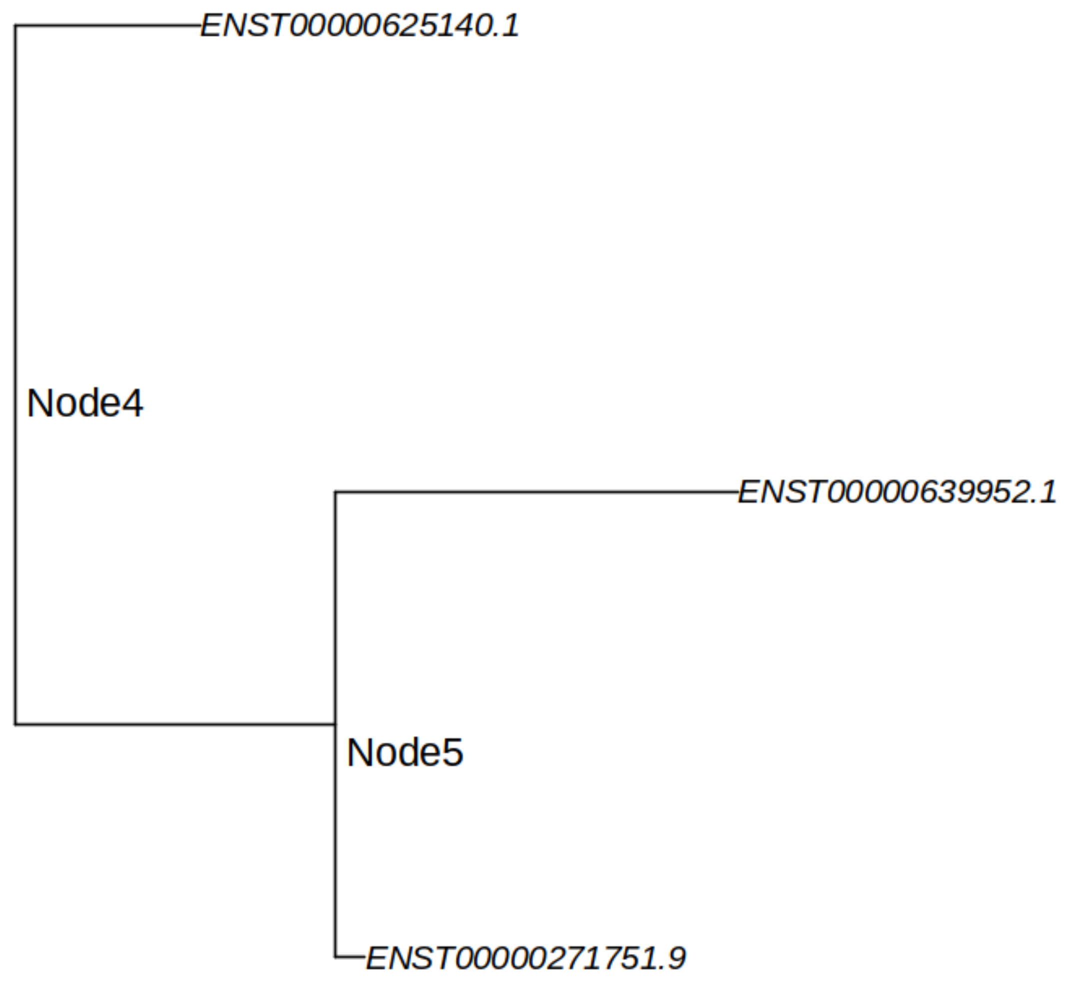
Discrete groups are not always optimal for downstream tasks.

Tree structure output by **TreeTerminus** can help in addressing these issues.



Building blocks

Construct a graph G = (V, E) where :

V = set of transcripts

E = $\{v_i, v_j\}$ such that v_i, v_j co-occur within at least one equivalence class

Motivation: Transcripts sharing *no* reads are unlikely to be good candidates for merging

Edge weight is $s(t_i, t_j)$

$$s(t_i, t_j) = \mathrm{infRV}_{i+j} - \frac{\mathrm{infRV}_i + \mathrm{infRV}_j}{2}$$

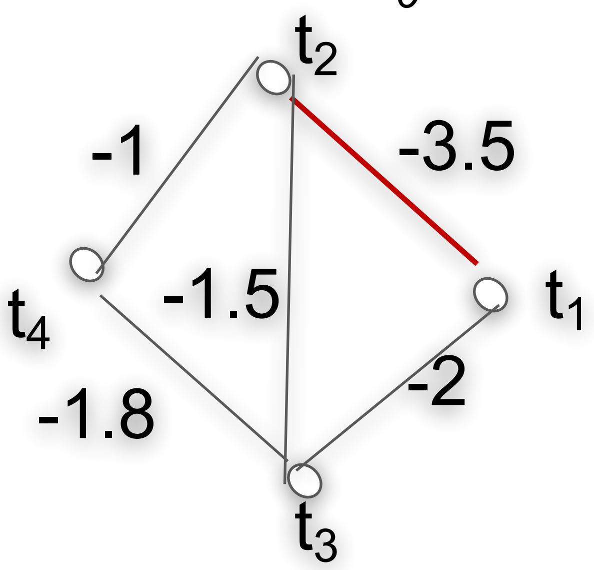
$$\mathrm{infRV}_i = \frac{\max(\sigma^2_{pi} - \mu_i, 0)}{\mu_i + pc} + d$$

The lower the edge weight, the better are those candidates for grouping.

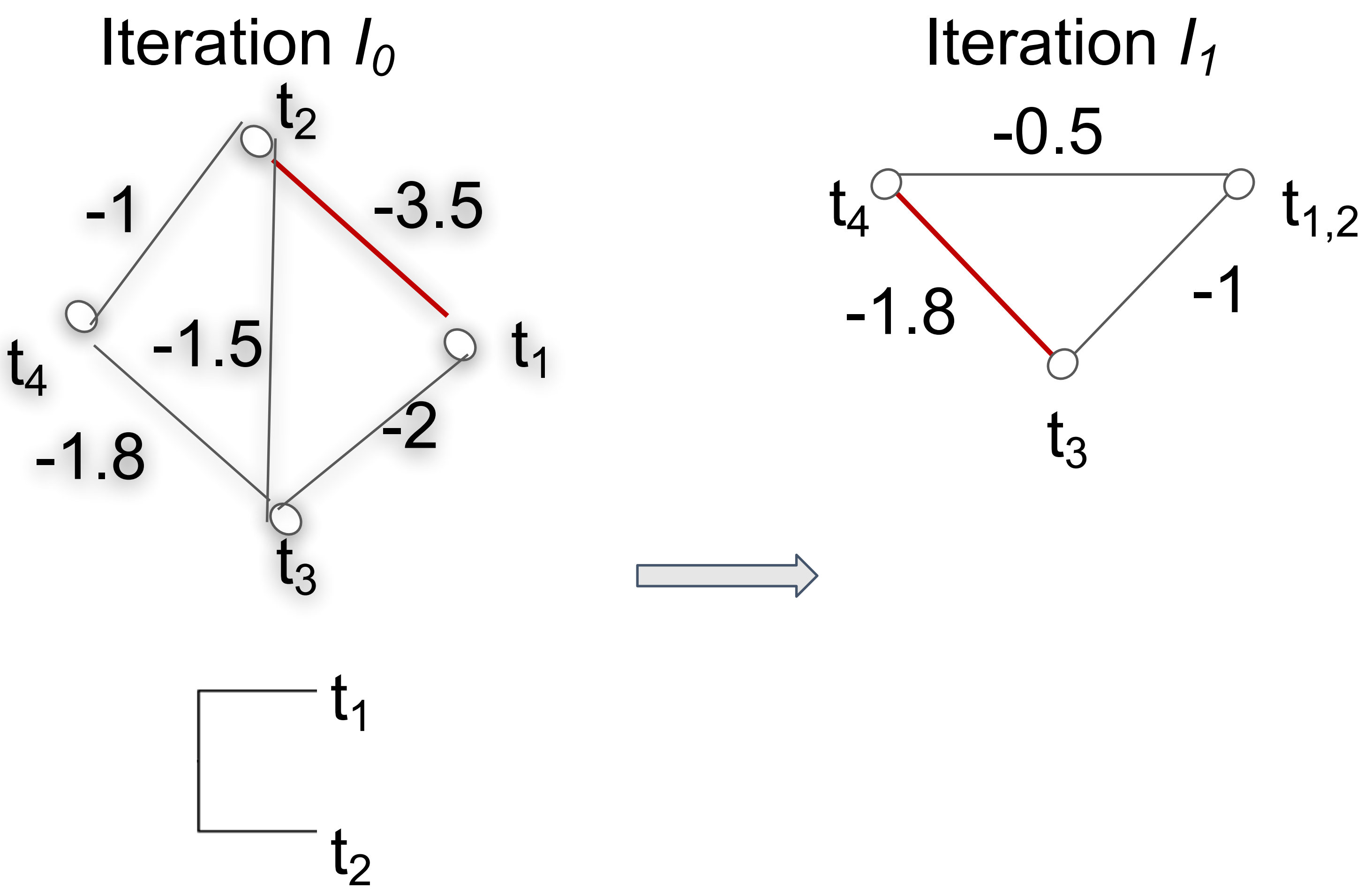
The graph G serves as an input for generating the forest for the procedure B(G).

Generating forest from the graph ($B(G)$)

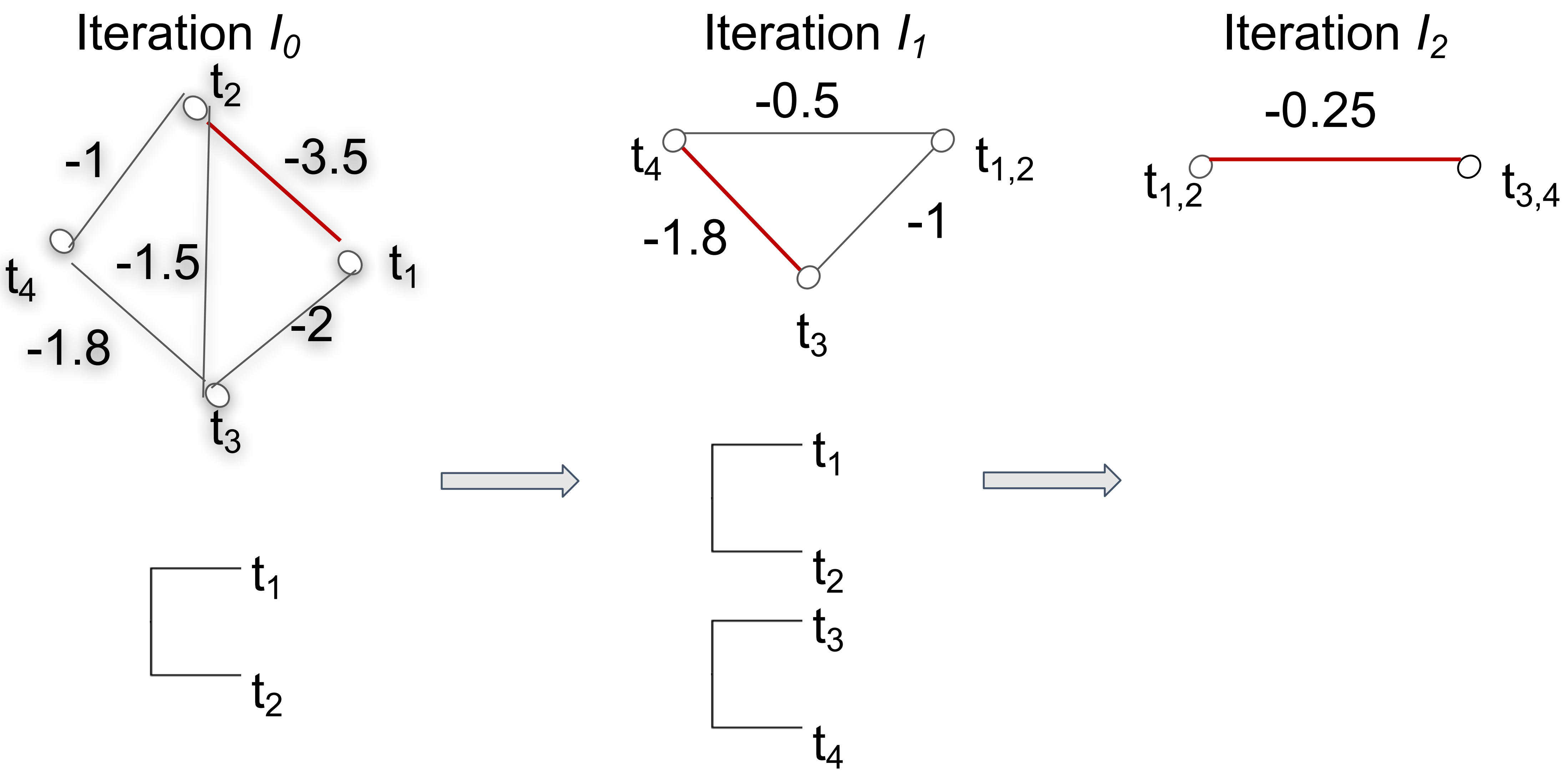
Iteration I_0



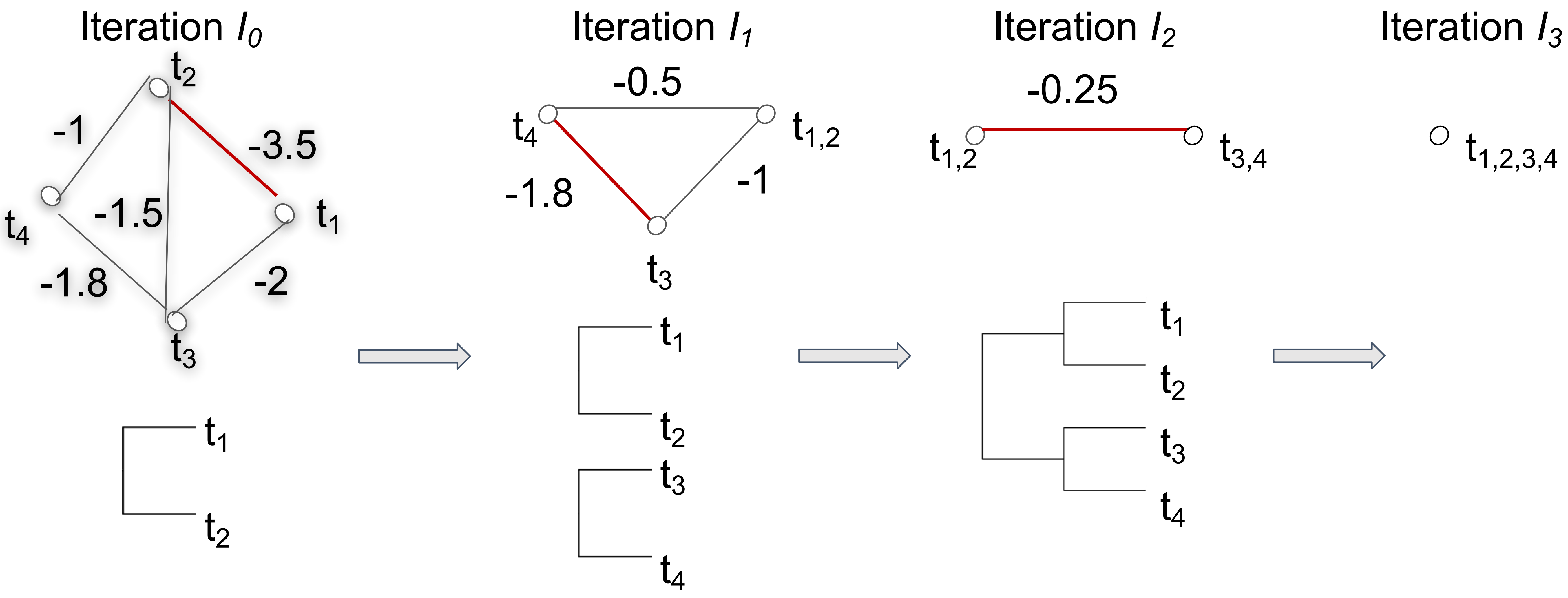
Generating forest from the graph ($B(G)$)



Generating forest from the graph (*B(G)*)



Generating forest from the graph ($B(G)$)



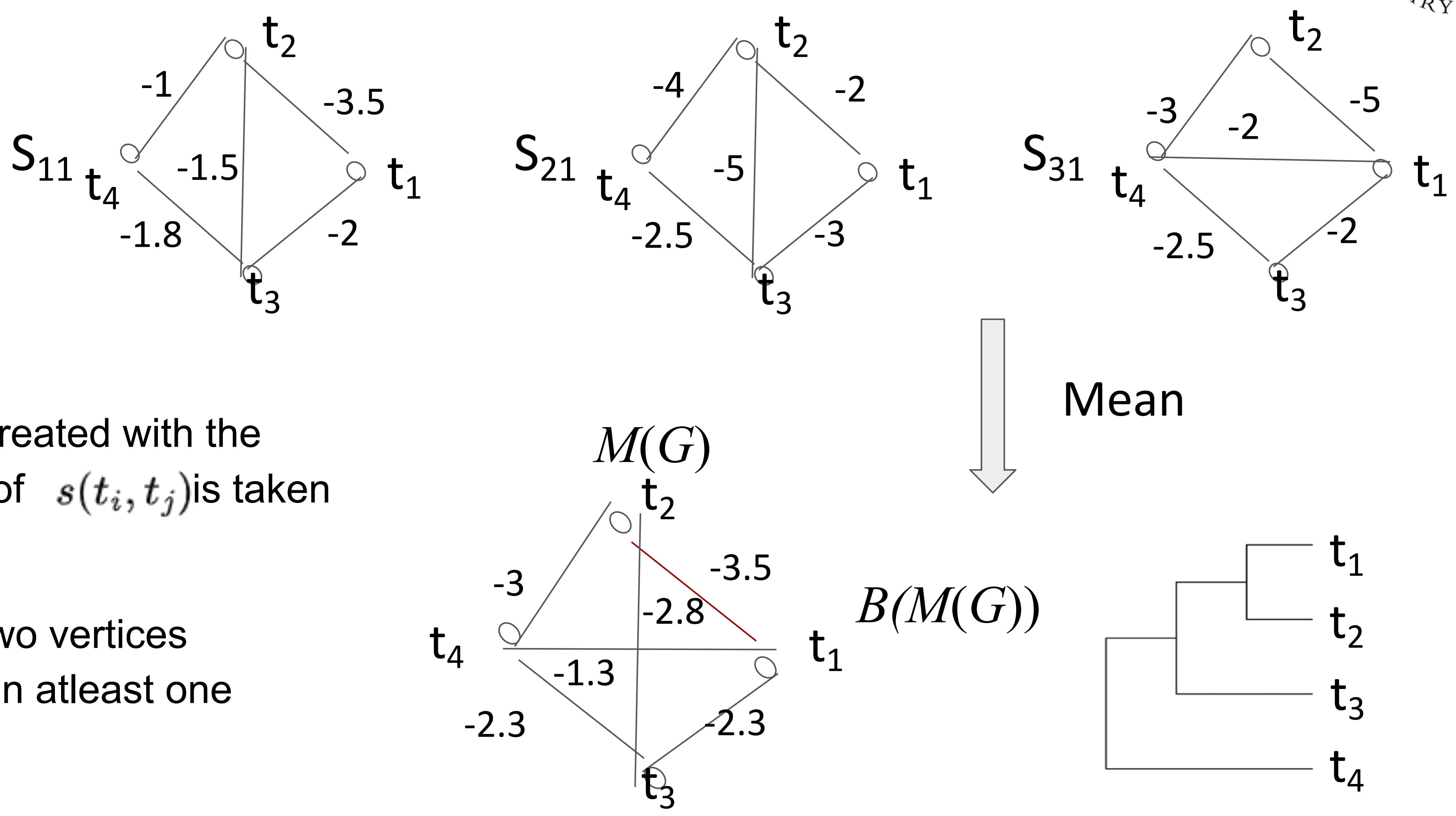
Generating forest across multiple samples

Generate a forest where the inferential uncertainty decreases across samples as we ascend the tree topology.

TreeTerminus provides two modes for generating the forest of trees for an RNA-Seq experiment.

- Mean
- Consensus

Mean Tree

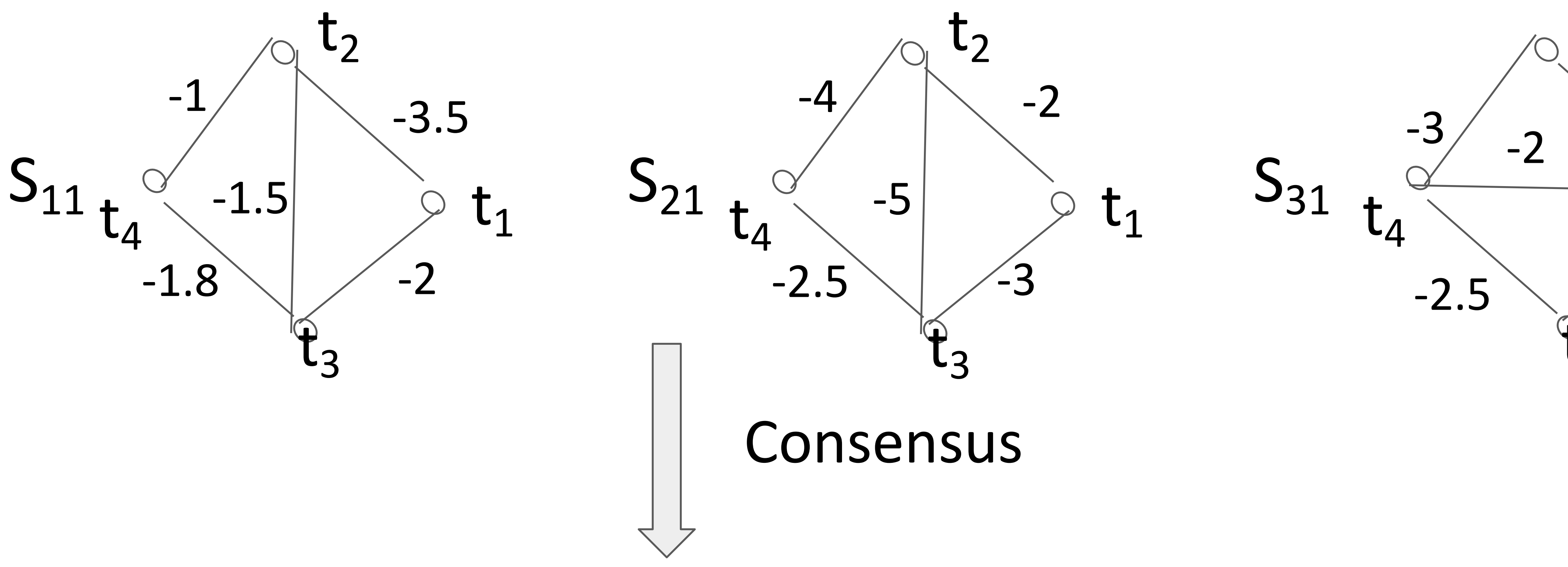


A new graph ($M(G)$) is created with the weight being the mean of $s(t_i, t_j)$ is taken across all samples.

An edge between any two vertices indicates that it existed in atleast one sample graph.

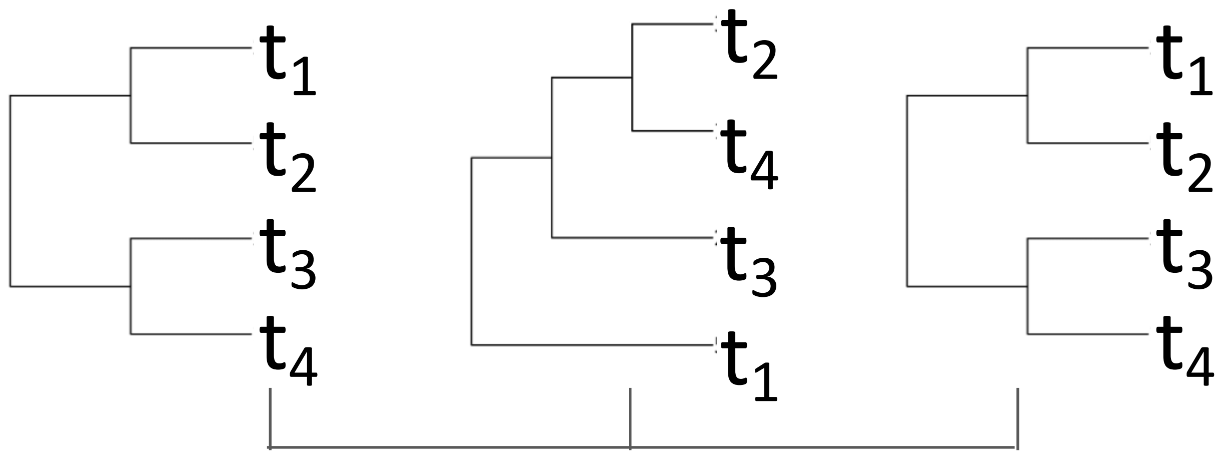
Procedure B is then applied on $M(G)$ to get the final forest of tree(s).

Consensus

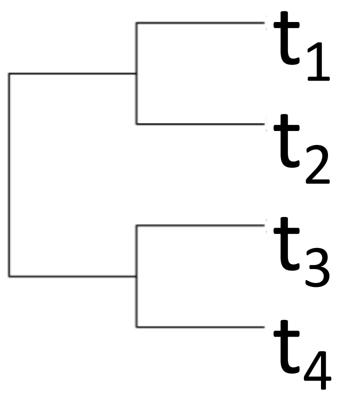


Consensus

$B(G_1)_1 = T_{11}$ $B(G_2)_1 = T_{21}$ $B(G_3)_1 = T_{31}$



Majority Rule Extended Consensus Algorithm

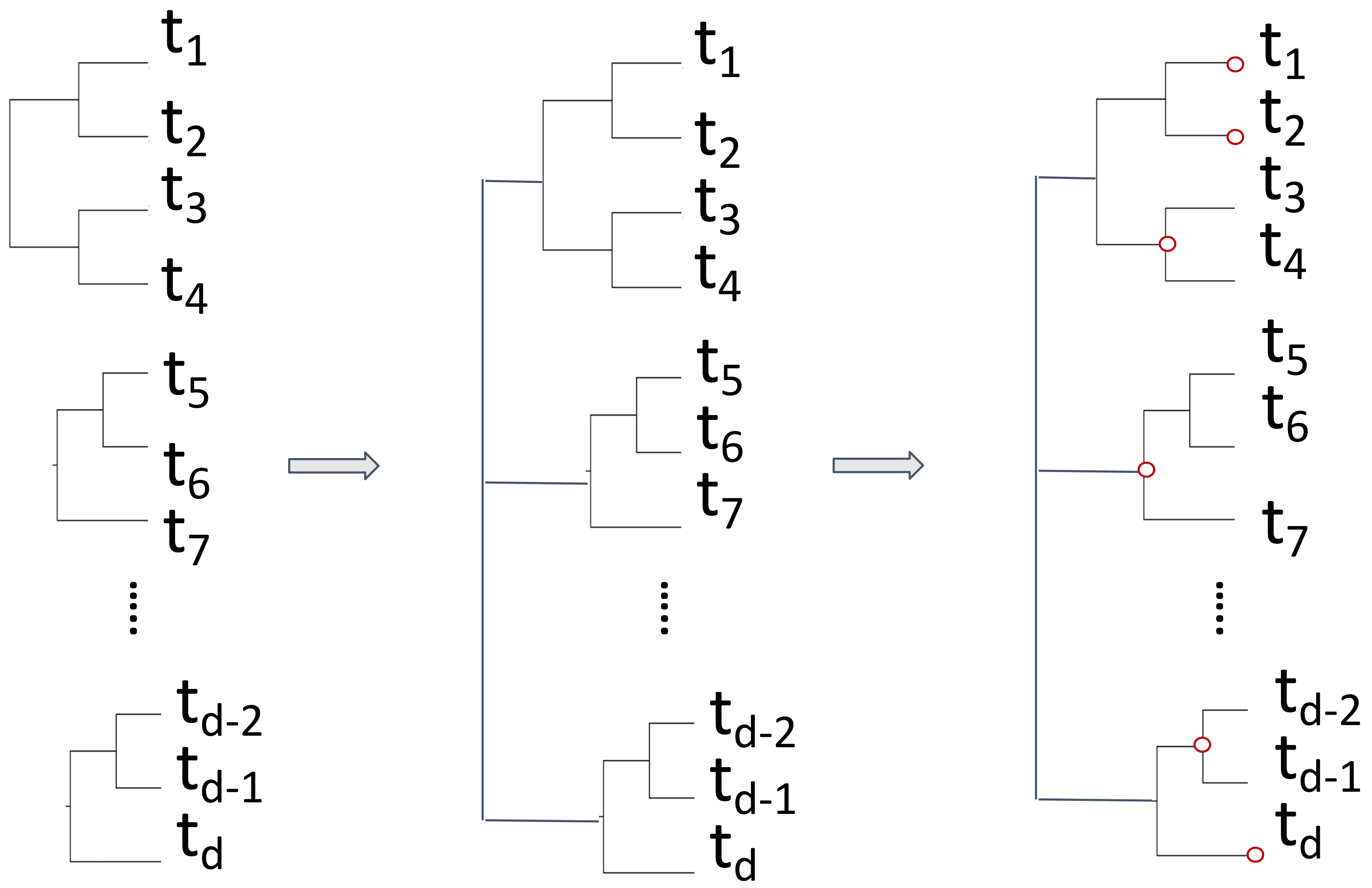


Fixed groups from TreeTerminus for downstream analysis

Forest output by TreeTerminus

Unified Tree

Find a cut that optimizes an objective function



Dynamic programming (DP) to find a cut in the tree that optimizes objective functions of the following form:

$$\underset{\mathcal{C}}{\operatorname{argmin}}\, f(\mathcal{C}) \text{ or } \underset{\mathcal{C}}{\operatorname{argmax}}\, f(\mathcal{C})$$

$$\text{where } f(\mathcal{C}) = \sum_{c \in \mathcal{C}} \operatorname{metric}(c)$$

Cut

$\{t_1, t_2, (t_3,t_4), (t_5,t_6,t_7), \ldots, (t_{d-2},t_{d-1}), t_d\}$

Objective Functions

Minimize weighted mean infRV and height - A cut which contains nodes that are situated at a low height in the tree but at the same time have lower uncertainty.

$$\underset{\mathcal{C}}{\operatorname{argmin}} \sum_{c \in \mathcal{C}} \mathrm{irv\_height\_desc}(c)$$

$$\mathrm{irv\_height\_desc}(c) = \mathrm{irv\_height}(c) * |\Lambda(c)|$$

$$\mathrm{irv\_height}(c) = \mathrm{mirv}(c) + \gamma\, \mathrm{height}(c)$$

$$\underset{\mathcal{C}}{\operatorname{argmax}} \sum_{c \in \mathcal{C}} \mathrm{wlfc\_desc}(c)$$

$$\mathrm{wlfc\_desc}(c) = \mathrm{wlfc}(c) \cdot |\Lambda(c)|$$

$$\mathrm{wlfc(c)} = \frac{||\mathrm{lfc(c)}||}{\mathrm{mirv(c)}}$$

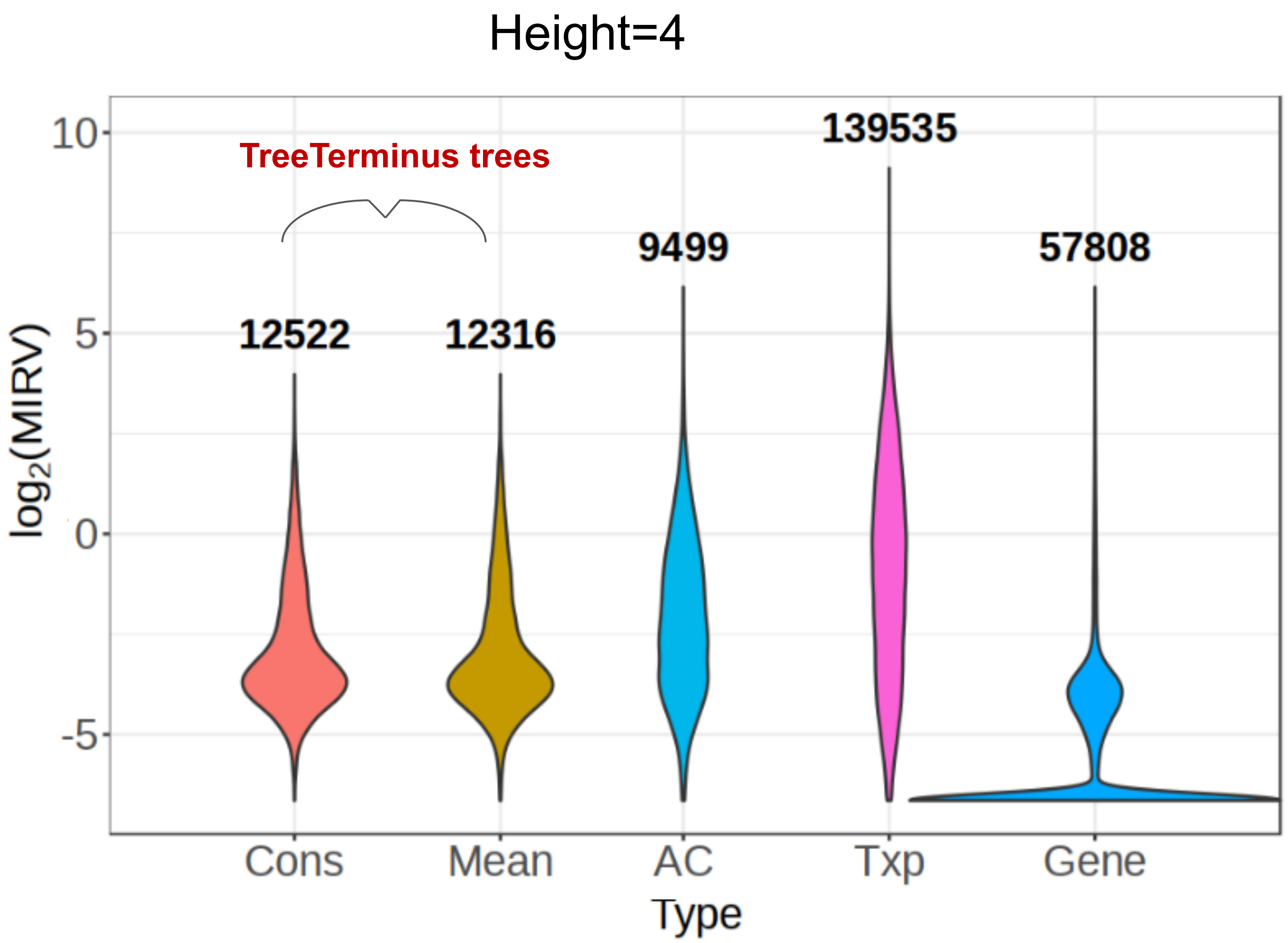
Maximize lfc weighted by mean infRV - A cut which contains nodes that have a high log fold change but at the same time have lower uncertainty.

Anti-correlation tree to serve as baseline

- No other methods output transcript-trees.
- Taking motivation from `mmcollapse`, we use the negative or the anti-correlation between a pair of transcripts computed on the inferential replicates as the metric for grouping.
- We use UPGMA for constructing the tree and pass it as an input to the distance matrix created from the anti-correlation matrix.

TreeTerminus tree nodes have low mean infRV

Simulated Dataset



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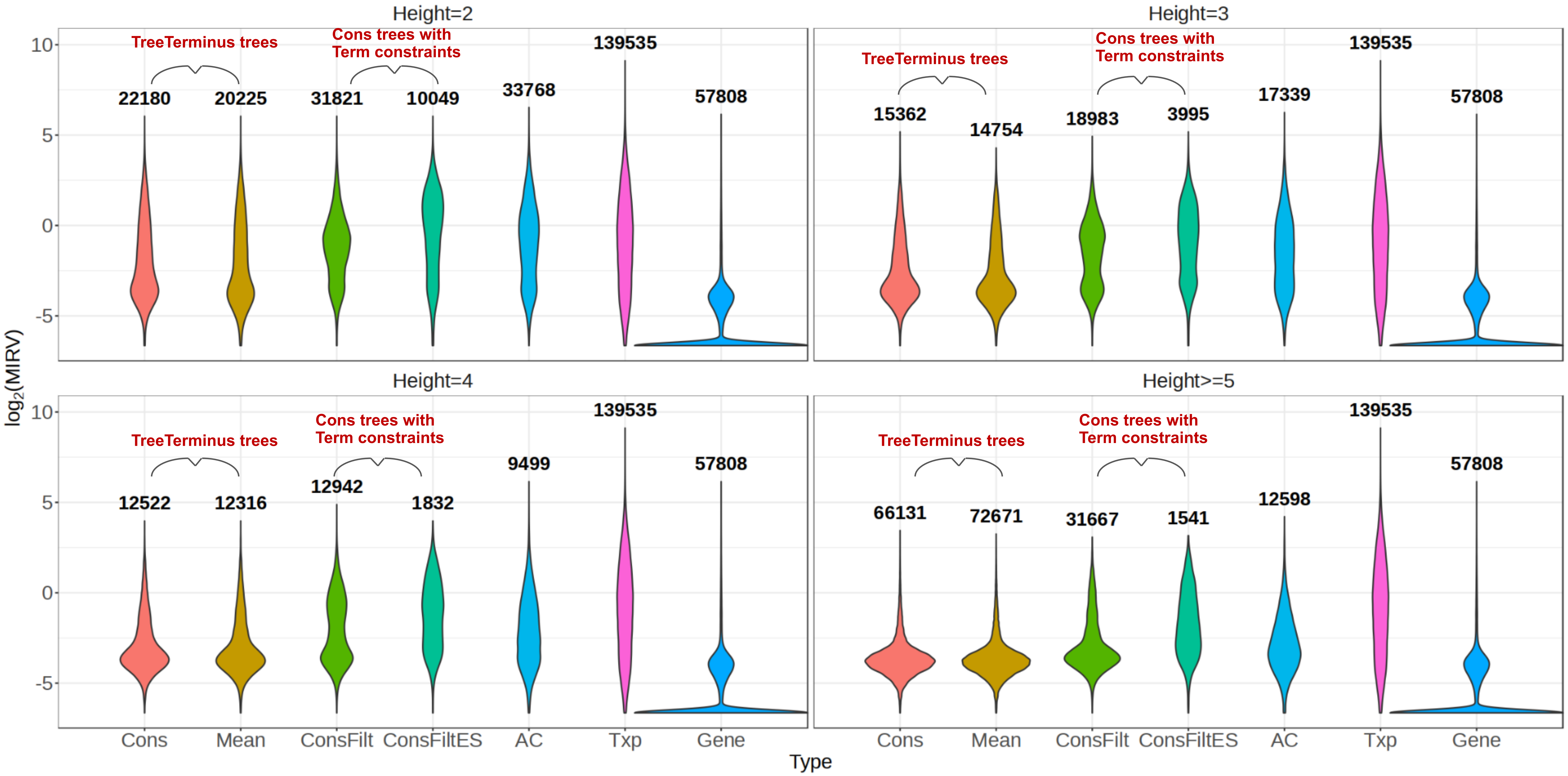
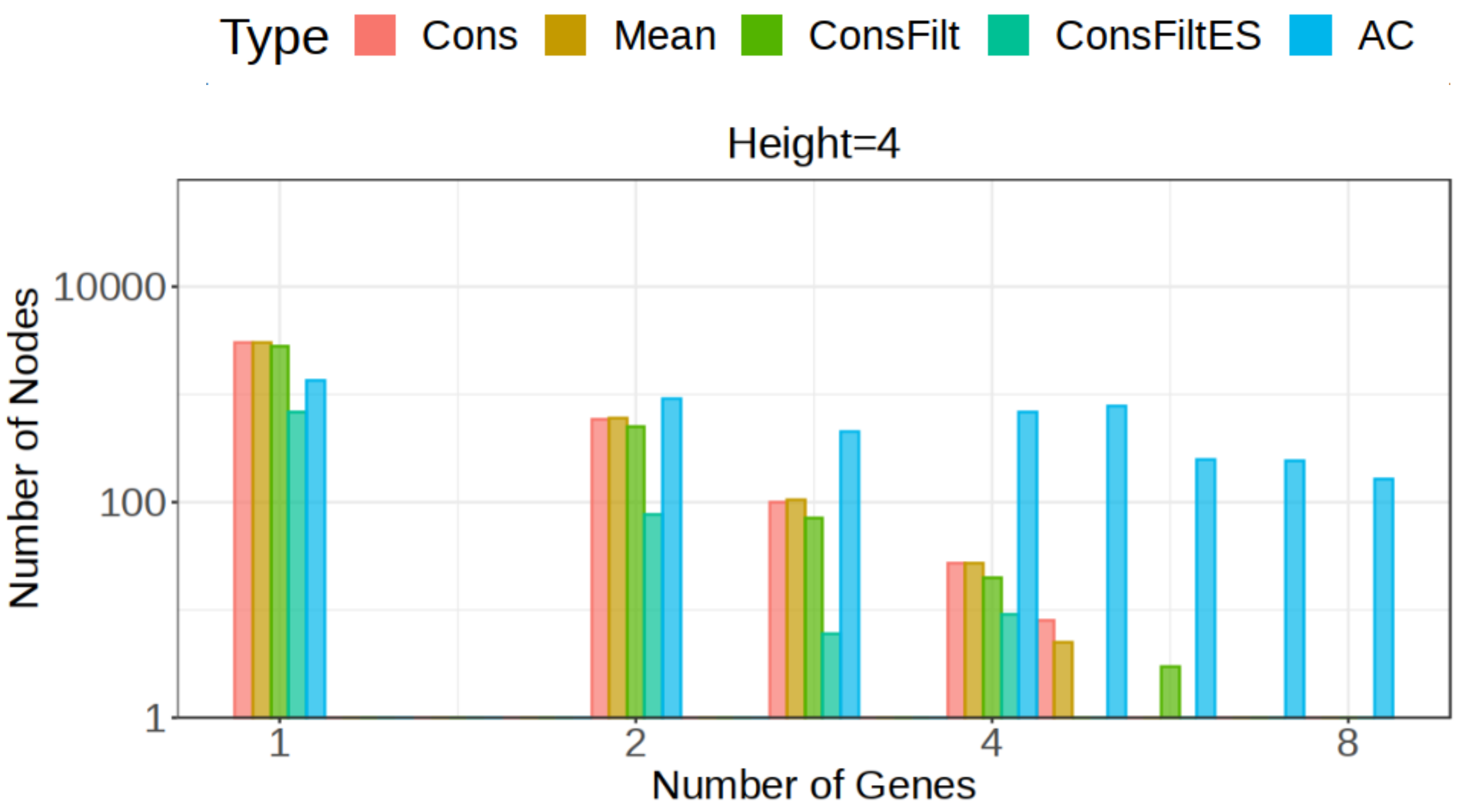


TreeTerminus nodes map to relatively fewer genes

ChimpBrain Dataset



Optimal performance is observed for differential expression analysis on the cuts obtained from TreeTerminus trees

Simulated Brain Dataset

	FDR			TPR		
	0.01	**0.05**	**0.1**	**0.01**	**0.05**	**0.1**
Obj[mirv(Ɣ=5)]	0.007	0.040	0.083	0.220	0.356	0.416
Obj[lfc]	0.006	0.037	0.081	**0.314**	**0.437**	**0.495**
Gene	**0.002**	**0.025**	**0.061**	0.256	0.413	0.491
Txp	0.008	0.039	0.080	0.208	0.345	0.407
Term	0.008	0.038	0.079	0.223	0.352	0.415

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Nodes belonging to `Obj[lfc]` cut contain more than 2200 true positive transcripts that are neither covered by Terminus groups or transcripts.

On the other hand, there are only 278 true positive transcripts that are unique to terminus but not covered by `Obj[lfc]` cut.

Summary

TreeTerminus

- A one of its kind method that outputs a forest of transcript-trees.
- Trees provide the end user with the flexibility to analyze data at different resolutions.
- TreeTerminus can be used to find the signal w.r.t transcripts using the inner nodes which would have been lost by doing transcript level analysis.
- End-user can design their own objective functions to find a suitable cut from the tree.

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Future Work

- Differential testing method that leverages the tree structure for RNA-Seq analysis.
- Application to droplet based single cell protocols, where the focus should be on grouping genes.

Differential Expression Analysis

We have talked about RNA-Seq quantification and how can we quantify and incorporate uncertainty to form transcript groups.

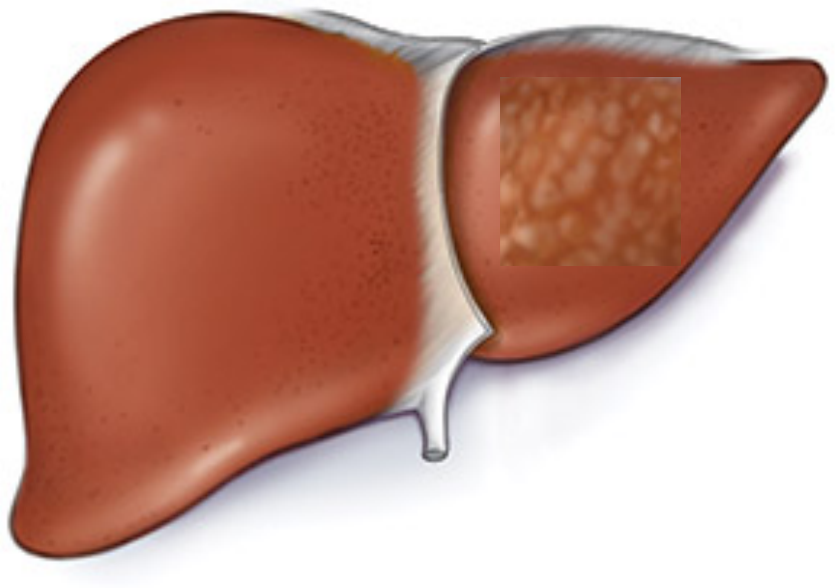
A question can arise is what to do with the quantified transcripts/features.

One most common application is to differential expression analysis

The goal of differential expression analysis is to identify transcripts/genes that are expressed at significantly different levels between two or more experimental conditions (e.g., disease vs. healthy, treated vs. untreated).

By comparing gene expression across conditions, we can gain insights into biological processes affected by the condition(s) of interest.

Case/control differential testing



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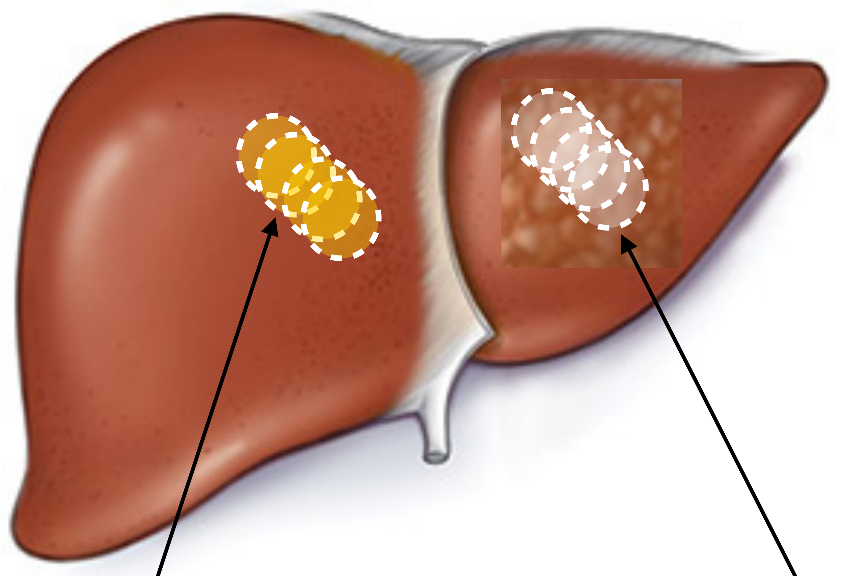


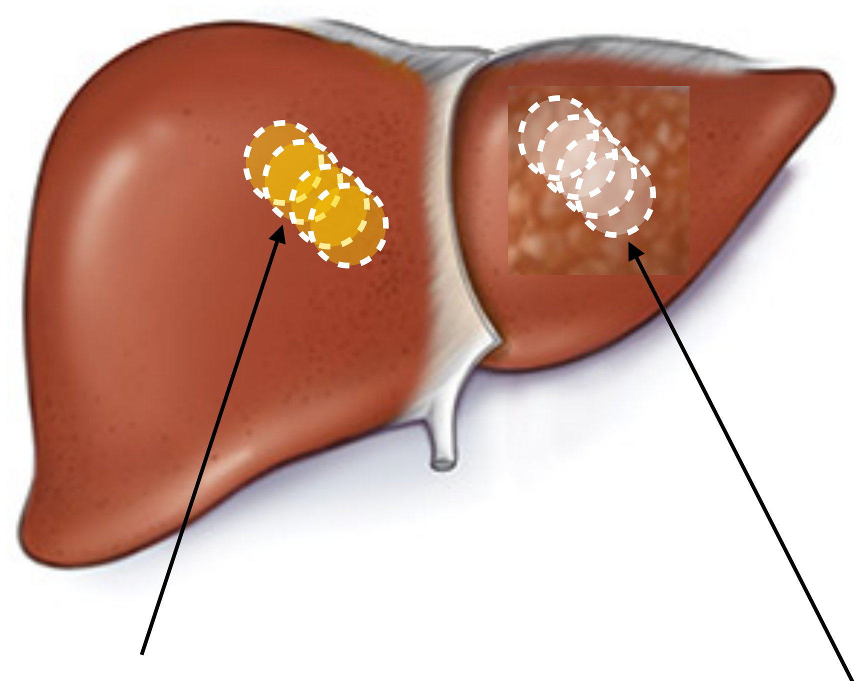
Control — healthy tissue

Case — diseased tissue

What is the difference in gene (transcript) expression profiles between these different samples?

Why multiple samples per “group”?

Case/control differential testing



What is the difference in gene (transcript) expression profiles between these different samples?

Why multiple samples per "group"?

> **Biological variability! Gene expression naturally varies "within" the same condition, we are mostly interested in expression changes whose magnitude is statistically significant w.r.t this background variability.**

Most methods are trying to improve over the (obvious) idea of doing a T-test between conditions (which is actually not too bad when you have a very large number of samples per condition)

Differential expression methods

DGE Tool	Publish Year
edgeR [34]	2010
Cuffdiff/Cuffdiff2 [35, 36]	2012/2013
DESeq2 [37]	2014
limma [38]	2015
DEGseq [39]	2009
baySeq [40]	2010
SAMseq [41]	2013
sleuth [42]	2017
NOIseq [43]	2012
Swish	2019

Non Parametric Methods

Recall Inferential Relative Variance

Inferential relative variance

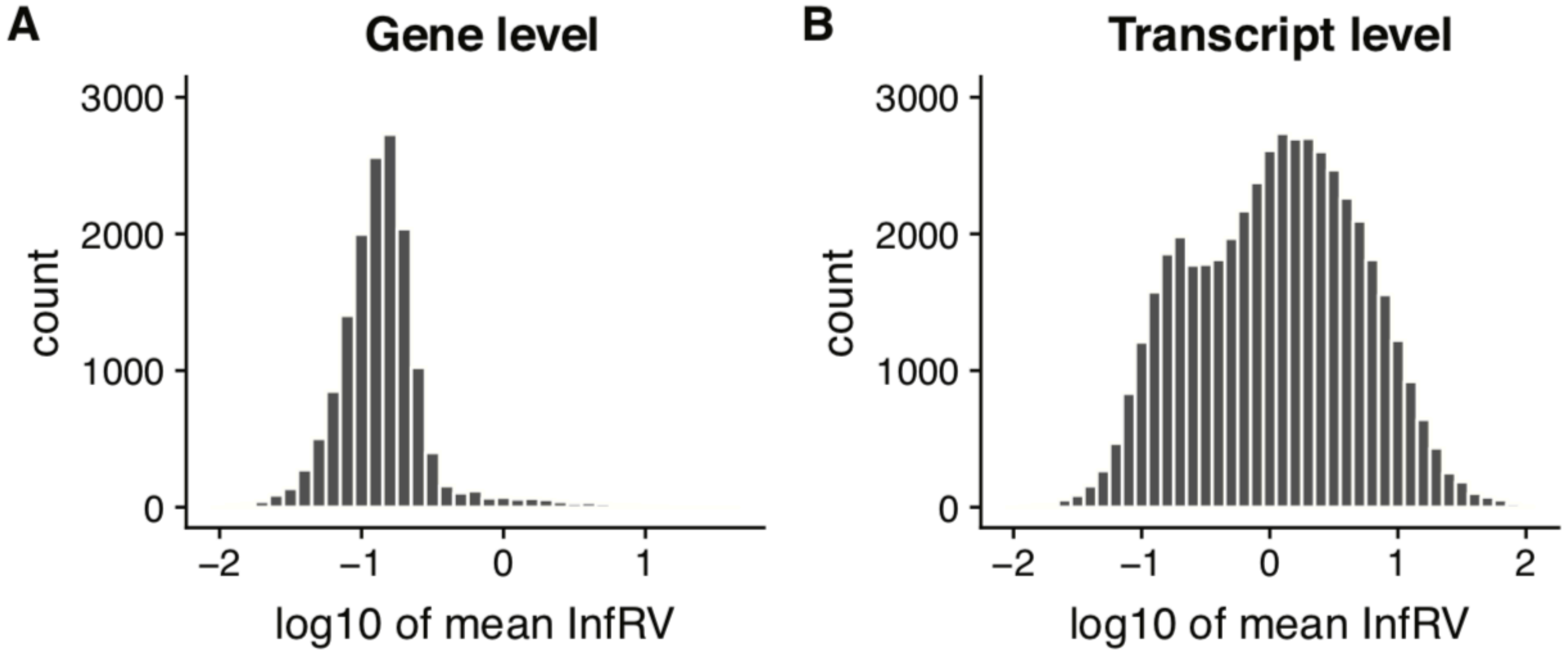
$$\mathrm{infRV}_{\mathbf{g_i}} = \frac{\max(0, \mathrm{var}(\mathbf{g_i}) - \mathrm{mean}(\mathbf{g_i}))}{\mathrm{mean}(\mathbf{g_i}) + p} + s$$

Why (variance - mean) / mean?

Minimally within a sample we have Poisson variance which scales with the mean.

So InfRV = ~0 with no multi-mapping.

Larger the value of InfRV, more uncertainty



Differential expression methods

Take Inferential uncertainty into account

DGE Tool	Publish Year
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NOIseq [43]	2012
Swish	2019

*The recent version of edgeR has this functionality

Swish

A non-parametric method that extends upon another non-parametric for RNA-Seq SAMSeq but also includes the inferential replicates

A non-parametric differential testing method does not assume that the underlying data follows a certain probabilistic distribution.

Depending on the nature of the tests, different ways to compute the test-statistic:

- Mann-Whitney U test
- Wilcoxon signed-rank test
- Kruskal-Wallis test
- Permutation test
- ....

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doi: 10.1093/nar/gkz622

Nonparametric expression analysis using inferential replicate counts

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[1]Department of Biostatistics, University of North Carolina-Chapel Hill, 135 Dauer Drive, Chapel Hill, NC 27599, USA, [2]Department of Computer Science, Stony Brook University, Computer Science Building, Engineering Dr, Stony Brook, NY 11794, USA and [3]Department of Genetics, University of North Carolina-Chapel Hill, 120 Mason Farm Rd, Chapel Hill, NC 27514, USA

Swish

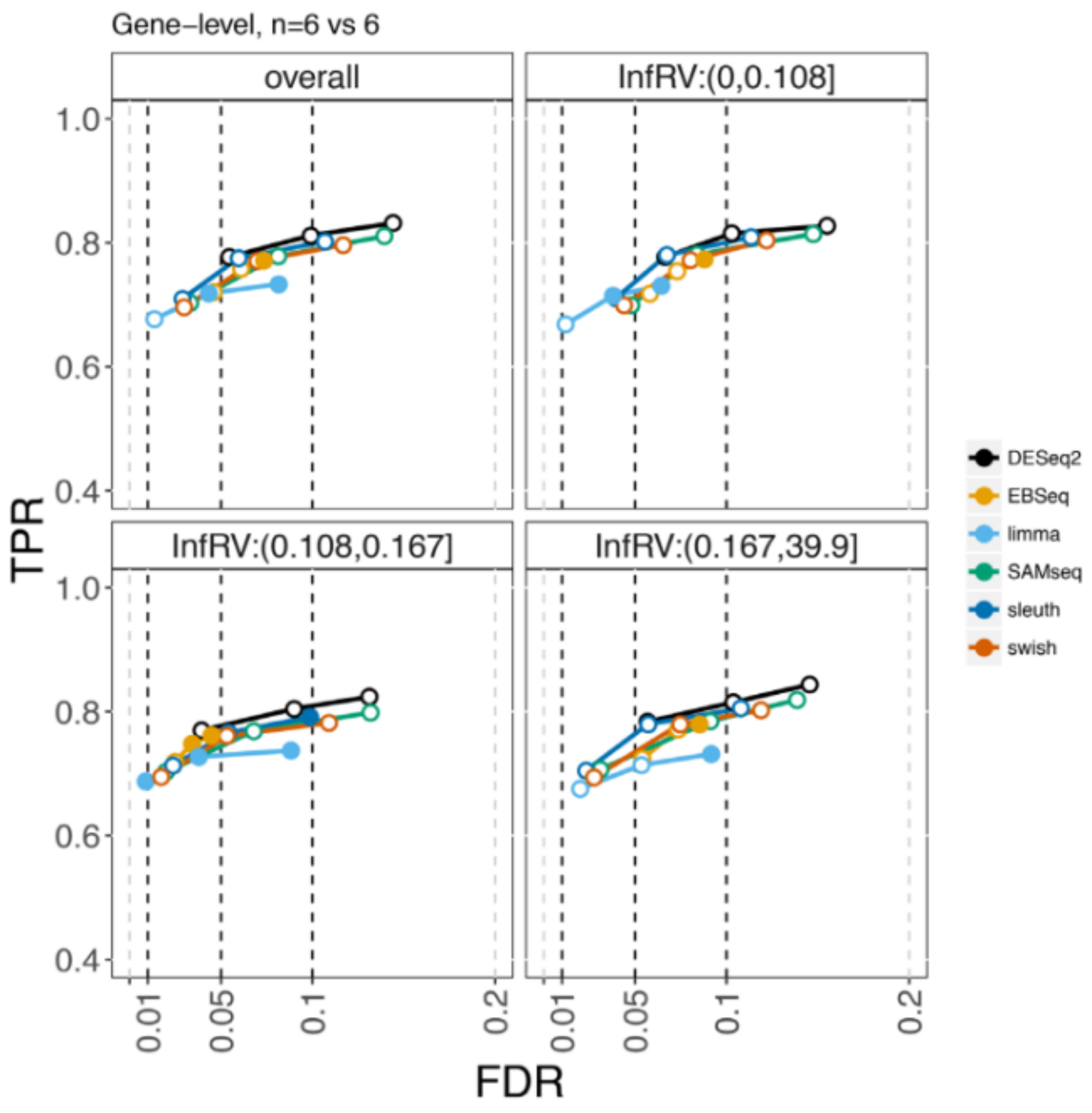
- Let there be m samples, with m_k denoting samples corresponding to condition k=$1,2$ and Y_{gi}^{s} be the scaled count from the Gibbs samples $(s = 1..S)$.
- If R_{gi}^{s} denotes the rank for the sample i in m biological replicates for gene g at the s gibbs sample, compute Mann-Whitney Wilcoxon test statistic $W_g^s = \sum_{i \in C_2} R_{gi}^s - \frac{m_2(m+1)}{2}$
- Under null hypothesis that gene g is not differentially expressed, W_g^{s} has expectation zero.
- For each gene g, final test statistic $T_g = \frac{1}{S} \sum_{s=1}^{S} W_g^s$ is the mean of W calculated over all Gibbs samples.
- This is very similar to SamSeq where mean is computed over downsampled poisson matrices rather than inferential replicates.

Estimating p-value

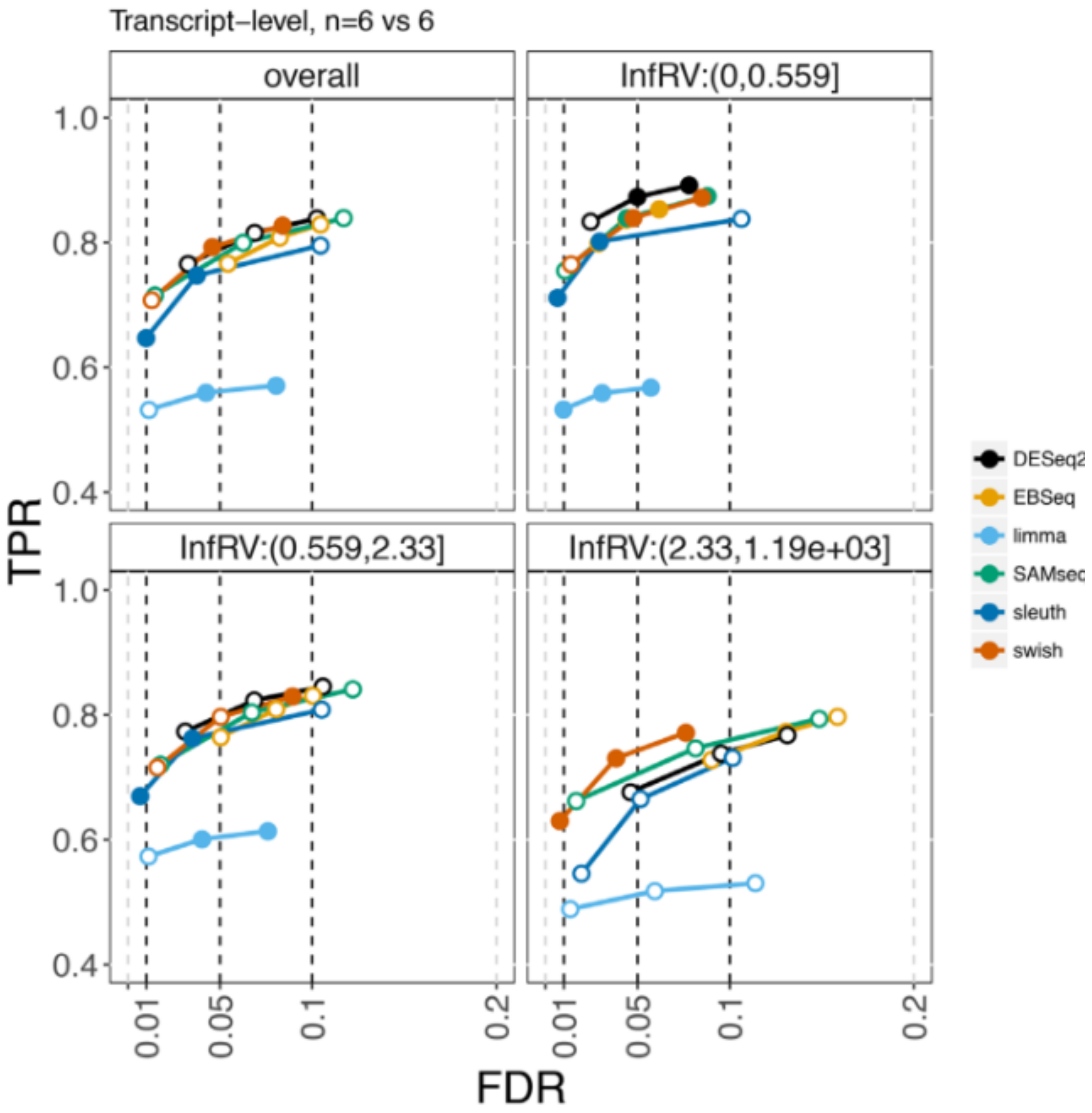
- The null distribution for T_g is not known and thus permutation testing is employed to compute the pvalue.

- Permute the samples B times and compute $T_g^{\,b}$ based on permuted data (b in $1..B$). p-value is then computed as proportion of how many times T_g is larger than T_i^q (b in $1..G$ and q in $1..B$).

Evaluating DGE



Evaluating DTE



Be careful of parametric assumptions as sample sizes increase